Human TFEB in HeLa Cells: Controlled Replication and a Test of C-Terminal Regulation

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Abstract

Transcription factor EB (TFEB) connects nutrient signaling to lysosomal and autophagy-related gene expression. This proposal combines controlled replication of TFEB redistribution with a hypothesis-driven investigation of C-terminal regulation in HeLa cells. The accepted WT reporter is 6,134 bp and encodes 722 residues, with a theoretical unmodified mass of 80.42 kDa. EGFP-only, S211A, S142A and S142A/S211A controls support matched starvation, Torin1 and refeeding experiments. Endogenous measurements, expression titration, pathway perturbation and TFEB loss/rescue address alternative explanations. A research extension compares WT, S466A/S467A/S469A (3A) and S466T/S467T/S469T (3T) with or without the linker–EGFP fusion under vehicle, Torin1 or MK-8722. Five additional sequence-defined constructs test architecture- and stimulus-dependent effects on target-gene RNA. Nuclear exposure, RNA production versus stability, and TFE3 compensation constrain mechanistic interpretation. Phosphosite and promoter-occupancy assays are conditional follow-ups. Independent culture-day replication, coded analysis and a prespecified interaction contrast govern inference. DNA calculations and maps are in-silico outputs; no cellular outcome is presented as measured data.

Keywords: TFEB; EGFP; phosphorylation; experimental controls; nuclear localization; autophagic turnover

1 Rationale and Research Basis

TFEB is a member of the MiT/TFE transcription-factor family and a major transcriptional regulator of the autophagy–lysosome pathway. Under nutrient-rich conditions, mTORC1-dependent phosphorylation favors cytoplasmic retention of TFEB. During nutrient starvation or mTOR inhibition, TFEB accumulates in the nucleus and activates genes involved in lysosomal biogenesis and cellular clearance [33, 34]. Nutrient-dependent control of nuclear export also contributes to TFEB distribution [12].

HeLa cells expressing TFEB–GFP have already been used to measure nuclear accumulation after starvation or Torin1 treatment [33]. The present course project applies this established biological response to evaluate a newly designed reporter; it does not claim discovery of the mechanism. Its primary readout is the nuclear-to-cytoplasmic distribution of the fusion protein; additional controlled modules test reversibility, endogenous consistency and downstream function. Molecular identity must be established before that distribution can be interpreted.

1.1 Research Objectives and Hypothesis

The first objective is to construct and sequence-verify an EGFP-tagged TFEB mammalian expression plasmid. The second is to test nutrient- and drug-dependent redistribution and recovery using reporter and endogenous measurements. The third is to assess selected phospho-site contributions and TFEB-dependent transcriptional and turnover responses. The replication hypothesis is that starvation increases the nuclear-to-cytoplasmic fluorescence ratio relative to time-matched nutrient-replete culture, and that Torin1 increases the ratio relative to its matched vehicle control. A separate research objective tests whether C-terminal S466/S467/S469 substitution effects depend on the EGFP fusion and stimulus. That extension is motivated by unresolved interpretation of this cluster and uses its own factorial design and decision rules.

2 Experimental Design

2.1 Overall Workflow

The study proceeds from reference-sequence retrieval and primer design to PCR amplification, plasmid construction, clone verification, transient transfection, and localization analysis. Sequence verification is the gate between molecular cloning and cell-based experiments.

2.2 Sequence-Defined Construct and Materials

The proposed construct, pEGFP-N1-TFEB-Bv1, will express human TFEB followed by a C-terminal EGFP tag under the CMV promoter. The reference will be TFEB transcript variant 1, NM_007162.3, whose CDS at transcript positions 303–1733 encodes NP_009093.1 (476 amino acids). A native, intron-free 1,431-nt CDS containing its original `TGA` will be ordered as linear double-stranded DNA without codon optimization, for example through the gBlocks HiFi synthesis service. This defined template will avoid uncertainty about the TFEB isoform in a cellular cDNA preparation. The synthesis feasibility, supplier QC and physical material identity will be checked before use; physical template and vector stocks have not been experimentally verified in this project [9, 13].

The empty backbone will be pEGFP-N1, GenBank U55762.1 (4,733 bp). Its bacterial kanamycin resistance and mammalian neomycin/G418 selection have different uses. The reference map includes the CMV promoter, EGFP, polyadenylation signals and replication origins, but the manufacturer's reference sequence does not certify a physical stock [3, 8]. C-terminal EGFP has a precedent in TFEB localization experiments; the present HindIII/BamHI design is not claimed to be sequence-identical to Addgene #38119 or its historical HindIII/KpnI construction [7, 33].

The accepted *in silico* design is 6,134 bp (Figure 8). It retains the native TFEB start codon, removes the TFEB stop through reverse-primer placement, and connects TFEB to EGFP through the seven-residue `GDPPVAT` linker. EGFP's `ATG` remains an internal methionine codon, and its terminal `TAA` ends the 722-residue fusion. The forward primer adds `ACC` immediately before TFEB's start, giving an A at position -3 and G at $+4$. Reverse-primer `CC` contributes coding-strand `GG` to preserve the reading frame. Removing a stop alone would not establish this frame; both junctions and the entire fusion were checked computationally.

The proposed laboratory workflow (Figure 9) will use Q5 High-Fidelity $2\times$ Master Mix (NEB M0492) for cloning PCR, HindIII-HF R3104 and BamHI-HF R3136 in rCutSmart, Monarch cleanup and gel-extraction kits, T4 DNA Ligase M0202, NEB 5-alpha C2987 competent *E. coli* (a DH5α derivative),

Monarch T1110 minipreps and Sanger primer walking. OneTaq M0482 will be used only for screening; its products will not enter the construct.

2.3 Cloning Primers and High-Fidelity PCR

Table 1 gives the complete oligonucleotides and their functional parts. The forward annealing segment matches CDS positions 1–22; the reverse segment is the reverse complement of CDS positions 1407–1428, excluding the native stop. HindIII and BamHI have no internal sites in the selected TFEB coding segment and provide distinct 5′ cohesive ends. Six protective bases per primer are a conservative choice: the manufacturer's end-cleavage table assigns both enzymes its highest category at three flanking base pairs, but that category represents 50–100% cleavage under the stated oligonucleotide assay conditions, not guaranteed complete digestion of this amplicon [15, 16, 20].

Table 1: Complete cloning primers and sequence-derived properties. Every sequence is written 5′ to 3′. In the partitioned lines, the order is protective bases, recognition site, deliberate addition, and template-annealing segment.

Primer	Sequence and properties
TFEB-HindIII-F	`ACTGACAAGCTTACCATGGCGTCACGCATAGGGTTGC`
	Parts: `ACTGAC` | `AAGCTT` | `ACC` | `ATGGCGTCACGCATAGGGTTGC`
	Full: 37 nt, 54.054% GC; annealing segment: 22 nt, 59.091% GC, $T_m = 69.498\,^{\circ}\mathrm{C}$. Full-oligo $T_m = 77.238\,^{\circ}\mathrm{C}$, not used for initial annealing.
TFEB-BamHI-R	`TATGACGGATCCCCCAGCACATCGCCCTCCTCCATG`
	Parts: `TATGAC` | `GGATCC` | `CC` | `CAGCACATCGCCCTCCTCCATG`
	Full: 36 nt, 61.111% GC; annealing segment: 22 nt, 63.636% GC, $T_m = 69.050\,^{\circ}\mathrm{C}$. Full-oligo $T_m = 78.896\,^{\circ}\mathrm{C}$, not used for initial annealing.

The reported estimates use Primer3-py 2.4.0 with SantaLucia thermodynamics and salt correction, 50 mM monovalent ions, 2.0 mM Mg^{2+}, 0.8 mM total dNTP and 500 nM primer. The monovalent-ion concentration is a modeling assumption, not a claim about Q5's proprietary buffer. The complete annealing segments and their terminal 12-mers each have one exact site in the defined template. Full-oligo hairpin and dimer screening passed the prespecified $55\,^{\circ}\mathrm{C}$ cutoff. These checks do not establish transcriptome-wide specificity or experimental amplification performance [31]. Only template-complementary segments will guide initial PCR annealing, because the added tails do not bind the original template.

Reactions will be assembled on ice using Table 2. The 1 ng template input is a design target for the synthetic linear template; the working dilution will be calculated from measured concentration. A no-template control will replace the template with water [29].

Table 2: Proposed Q5 cloning PCR, 50.0 μL total. Stock targets are not measured concentrations.

Component	Volume	Final concentration or input
Q5 M0492 2× Master Mix	25.0 μL	1×; Mg^{2+} 2.0 mM; each dNTP 200 μM
TFEB-HindIII-F, 10 μM	2.5 μL	0.5 μM
TFEB-BamHI-R, 10 μM	2.5 μL	0.5 μM
Synthetic template, target 1 ng/μL	1.0 μL	1 ng total
Nuclease-free water	19.0 μL	To 50.0 μL

The initial program will be 98 °C for 30 s; 30 cycles of 98 °C for 10 s, 65 °C for 20 s and 72 °C for 45 s; then 72 °C for 2 min. The 45 s extension rounds the expected 1,457-bp product upward at 30 s/kb. Initial annealing at 65 °C and a 62/65/68 °C optimization gradient are design choices, not Q5-calculator outputs or measured optima [29].

2.4 Product Assessment, Restriction Digestion and Ligation

A 5 μL PCR aliquot will be analyzed on 1.0% agarose in 1× TAE with 1× loading dye, a compatible DNA stain and 0.5 μg NEB N3232 1 kb ladder. An initial field of 5 V/cm of electrode separation is a design choice; electrophoresis will stop after the relevant bands separate and before the tracking front exits. The ladder's 500/517 bp pair is expected to co-migrate. Uncut plasmids will not be sized by comparison with linear DNA, and illumination of DNA intended for cloning will be minimized [25].

A single 1,457-bp PCR product will be purified with Monarch Spin PCR & DNA Cleanup T1130: 20–100 μL sample, five sample volumes of prepared BZ, an S1A column, two 200 μL WZ washes, and 20 μL EY elution after a 1 min wait. Centrifugation will follow the current kit protocol at $16{,}000 \times g$ and room temperature. If additional bands occur, the target band will be excised and purified with T1120 using four gel-slice volumes of BY, dissolution near 50 °C, an S1A column, two 200 μL WZ washes and 20 μL EY elution. Alcohol-containing buffers will be prepared according to their current bottle labels. Recovered DNA will be measured; no yield or recovery fraction is assumed [21, 22].

PCR product and empty vector will be digested separately with the common recipe in Table 3, at 37 °C for 60 min. Both enzymes have full listed activity in rCutSmart; the proposed enzyme volumes contribute 2% glycerol. Actual DNA and water volumes will be recalculated from measured concentrations. Insufficient material will prompt repeat or scaled preparation. Uncut and single-digest analytical controls will be included when troubleshooting incomplete digestion. BamHI-HF cannot be heat-inactivated, so digestion products will be purified before ligation [15, 18, 20].

For direct preparative or diagnostic gel loading of unpurified BamHI-HF digests, SDS-containing Purple Gel Loading Dye B7024 (6×) will be diluted to 1× by adding one volume of dye to five volumes of sample. This will dissociate residual enzyme-bound DNA; the no-SDS B7025 product will not substitute for it. Dye will be added only to gel samples. If SYBR Safe or GelRed is used, staining will occur after electrophoresis as directed for B7024 [15, 19].

The 4,695-bp backbone will be gel-purified to separate residual circular vector and the excised 38-nt MCS. The digested insert will be purified with T1130. Its nominal retained length is 1,439 nt; the discarded top-strand terminal pieces are 7 and 11 nt. These short pieces and the 38-nt MCS will not be expected to resolve usefully on the proposed gel. Both retained fragments have equal strand lengths; their pydna duplex spans are 4,699 and 1,443 bases because of staggered ends. Gel sizes will use the nominal

top-strand cut-to-cut lengths, and molarity will use mean strand lengths, which equal those nominal lengths for the selected ends. Distinct 5′ `AGCT` and `GATC` overhangs will support directional ligation. Routine dephosphorylation will be omitted; vector-only background will be used to assess incomplete digestion or circular-vector carryover.

The initial ligation will contain 20 fmol backbone and 60 fmol insert (3:1 insert:vector). With mean strand length L, the mass conversion is

$$m\ (\mathrm{ng}) = n\ (\mathrm{fmol}) \times L\ (\mathrm{nt}) \times 0.00066.$$

The calculated inputs are 61.974 ng backbone and 56.9844 ng insert. Table 3 gives a practical recipe rounded to 0.01 μL. The unrounded total DNA concentration is 5.948 ng/μL, within the manufacturer's 1–10 ng/μL starting range. Fully thawed ATP-containing ligase buffer will be mixed before use. Ligation will proceed at 20–25 °C for 10 min, followed by 65 °C for 10 min. This last treatment terminates the ligation reaction; it does not replace purification of the preceding BamHI-HF digests. A matched vector-only ligation will replace the insert volume with water. No polynucleotide kinase is needed after the planned restriction digestion [17].

Table 3: Proposed digestion and ligation mixtures. DNA stock concentrations are preparation targets to be adjusted using measurements.

Component	Volume	Amount or final condition
Each separate digestion: 50.0 μL		
DNA, target 50 ng/μL	10.0 μL	500 ng
10× rCutSmart	5.0 μL	1×
HindIII-HF, 20 U/μL	1.0 μL	20 U
BamHI-HF, 20 U/μL	1.0 μL	20 U
Water	33.0 μL	To 50.0 μL
Ligation: 20.00 μL		
Backbone, target 20 ng/μL	3.10 μL	Approximately 20 fmol
Insert, target 10 ng/μL	5.70 μL	Approximately 60 fmol
10× T4 DNA Ligase Buffer	2.00 μL	1×
T4 DNA Ligase M0202, 400 U/μL	1.00 μL	400 U
Water	8.20 μL	To 20.00 μL

2.5 Transformation, Screening and Plasmid Preparation

For each transformation, 2 μL ligation will be added to 50 μL NEB 5-alpha C2987 cells. Cells will remain on ice for 30 min, undergo heat shock at 42 °C for 30 s, return to ice for 5 min, and receive 950 μL SOC. Recovery will proceed at 37 °C and 250 rpm for 60 min. Then 100 μL each of undiluted recovery and 1:10 and 1:100 SOC dilutions will be plated on LB-kanamycin and incubated at 37 °C overnight. Kanamycin will be 30 μg/mL in bacterial plates and liquid cultures; G418 will not be used for bacterial cloning [26, 27].

Vector-only ligation, a no-DNA control and the supplied pUC19 transformation control will be processed in parallel. pUC19 will be selected on ampicillin at 100 μg/mL. These controls will assess backbone background, contamination or selection failure, and competent-cell handling, respectively. The pUC19 control will not test kanamycin resistance or the ligation itself. No transformation efficiency or fraction of correct clones is assumed [26].

Eight isolated colonies will initially be screened while numbered patches are retained. Each 25 μL OneTaq M0482 reaction will receive trace colony material, 12.5 μL 2$\times$ mix, 0.5 μL of each 10 μM diagnostic primer and 11.5 μL water; final primer concentrations will be 0.2 μM. Colony lysis will be $94\,^{\circ}$C for 5 min, followed by 30 cycles of $94\,^{\circ}$C for 30 s, $58\,^{\circ}$C for 30 s and $68\,^{\circ}$C for 160 s, then $68\,^{\circ}$C for 5 min. The 160 s extension covers the longest modeled 2,608-bp product at at least 1 min/kb. An initial $58\,^{\circ}$C annealing step and $55/58/61\,^{\circ}$C gradient are design choices [28].

S-F1/S-R1 will flank the insert, while S-F2/S-R1 will combine an internal TFEB primer with a downstream vector primer to test orientation (sequences and sites in Table 4). Their model T_m values are $60.04/62.55\,^{\circ}$C and $66.09/62.55\,^{\circ}$C, respectively, using 50 mM monovalent ions, 1.8 mM Mg^{2+}, 0.8 mM total dNTP and 200 nM primer. This is separate from the Sanger-primer model below [31]. The correct-construct predictions are 2,605 bp for flanking PCR and 2,093 bp for orientation PCR; an empty vector predicts 1,204 bp for flanking PCR and no specific orientation product. Empty-vector and no-template controls will accompany screening. A missing orientation band will remain provisional without a functioning positive control: in the first construction round, flanking-positive minipreps will proceed to sequencing if all orientation assays are negative. A sequence-accepted clone may provide the positive control in later rounds.

Up to three screening-positive colonies will be grown independently in 3 mL LB-kanamycin at $37\,^{\circ}$C for 12–16 h with aeration; if fewer are recovered, those available will be analyzed. For each T1110 miniprep, 1.5 mL culture will be processed with 200 μL RNase-containing B1, 200 μL B2 for 1 min with gentle inversion, then 400 μL B3 for 2 min. After clarification, DNA will bind to an S2D column, receive 200 μL BZ and 400 μL WZ washes, and elute in at least 30 μL EY. For DNA intended for transfection, the manufacturer's additional 5 min BZ incubation will be used; this is not an endotoxin-free certification. Yield and purity will be measured [23, 24].

A 500-ng miniprep aliquot will undergo the defined 50-μL HindIII/BamHI digestion and gel analysis with the B7024 handling above. Gross agreement with 4,695- and 1,439-bp fragments will support the insert/backbone sizes, but will not exclude reversed insert models, short indels or substitutions. A three-base retained-stop difference will not be resolved reliably on the proposed gel. Complete target sequencing will be the acceptance gate.

2.6 Sanger Primer Walking and Acceptance Criteria

Each candidate will be sequenced with the ten primers in Table 4, in separate reactions. For Eurofins Standard Read purified-plasmid submissions, this construct falls in the 150–250 ng/μL stock range. A target of 200 ng/μL and 55 μL DNA per clone will support ten reads in the separate-template format ($10 + 9 \times 5$ μL). If needed, independently prepared aliquots from the same single-colony culture will be pooled and measured or concentrated; different clones will not be pooled. Each primer will be supplied separately at 10 μM, with at least 10 μL for its first read and 5 μL per additional use. These are submission targets, not claimed miniprep yields; current service requirements will be rechecked before ordering [4].

Sanger-primer estimates use Primer3/SantaLucia with 50 mM monovalent ions, no added Mg^{2+} or dNTP, and 50 nM oligo. This reproducible comparison is not the sequencing service's proprietary chemistry. Selection required 18–25 nt, 35–60% GC, T_m of 50–$62\,^{\circ}$C, no four-base homopolymer, a terminal G/C without three consecutive terminal G/C bases, and modeled hairpin/homodimer $T_m < 40\,^{\circ}$C. Each oligo has one exact full-length binding site in the final construct; genome-wide specificity has not been tested [4, 31].

Table 4: Sanger and diagnostic primers. Coordinates are 1-based inclusive on pEGFP-N1-TFEB-Bv1. A plus/minus sign denotes the extension direction; every oligonucleotide is written 5′ to 3′. Windows are predictions, not measured reads.

ID	Complete sequence	nt; GC%; T_m (°C)	Binding site (strand)	Planned usable window
S-F1	GGGATTTCCAAGTCTCCACC	20; 55.00; 52.03	418–437 (+)	488–1087
S-F2	TGGGAACAAGTTTGCTGCCCAC	22; 54.55; 58.13	930–951 (+)	1002–1601
S-F3	ACCATCCTCAAGGCCTCTGTGG	22; 59.09; 57.96	1456–1477 (+)	1528–2127
S-F4	TGATCCACTTCTGTCCACCATGTCC	25; 52.00; 58.23	1971–1995 (+)	2046–2645
S-F5	GCTGGAGTACAACTACAACAGCCAC	25; 52.00; 57.83	2502–2526 (+)	2577–3176
S-R1	GACAAACCACAACTAGAATGCAGTG	25; 44.00; 54.94	2998–3022 (−)	2348–2947
S-R2	GTTGCCGTCCTCCTTGAAGTCG	22; 59.09; 57.87	2466–2487 (−)	1816–2415
S-R3	ATGGTGGACAGAAGTGGATCAGAGG	25; 52.00; 57.76	1967–1991 (−)	1317–1916
S-R4	ATCCACAGAGGCCTTGAGGATGG	23; 56.52; 57.86	1457–1479 (−)	807–1406
S-R5	TGATGTGGGCAGCAAACTTGTTCC	24; 50.00; 58.35	932–955 (−)	282–881

The conservative planning window comprises extended bases 51–650 after each primer's 3′ end (600 usable bases), excluding the early low-quality region and remaining within a standard-read length. Adjacent planned windows overlap by at least 50 nt. The acceptance target is positions 602–2799 (2,198 nt): 20 upstream vector bases, the 5′ junction and initiation context, and the full 2,169-nt fusion ORF through its final stop. The computational coverage check gives every target base at least one forward and one reverse read; primer-binding bases within this target are covered by other primers. The outer S-F1 and S-R1 sites lie outside the acceptance target. This plan does not establish bidirectional sequencing of those external sites or of the complete vector backbone [4, 6].

A clone will be accepted only after actual quality-trimmed chromatograms provide unambiguous, concordant Q20-or-better calls in both directions across the entire target, no unexpected nucleotide changes, and the predicted restriction pattern. Reads will be aligned to the exact B-v1 reference, with manual inspection of junctions and all mismatches. Q20 is a per-base error-probability metric, not a guarantee that a molecule is error-free [5]. Low-quality regions, single-strand gaps and ambiguous peaks will trigger additional reads or primers. Reproducible mixed peaks will prompt single-colony re-isolation and a fresh preparation. A clear unwanted coding or junction mutation will reject that clone; a shared error across independent clones will prompt investigation of template and design. Predicted coverage will never substitute for observed trace quality.

2.7 Troubleshooting and Experimental Boundary

No cloning-PCR band will prompt checks of template identity, dilution, primer matching and annealing conditions; additional bands will prompt the planned gradient and target-band purification. High vector-only background will prompt checks for incomplete digestion and residual circular DNA. Absence of colonies in both the test and positive transformation control will point to handling or selection problems before failed ligation is inferred. An apparently correct gel pattern with sequence disagreement will fail acceptance. Only sequence-accepted plasmid will enter the proposed HeLa experiment; expression, localization and biological behavior will still require cellular validation.

2.8 Sequence-Defined Control and Mutant Constructs

Five plasmid conditions are defined: parental pEGFP-N1 expressing EGFP only, the accepted WT TFEB–EGFP reporter, and three derived reporters carrying S211A, S142A or S142A/S211A (Figures 1 and 2). Amino-acid positions refer to human NP_009093.1. The fusion constructs share the same 6,134-bp plasmid architecture, the GDPPVAT junction and C-terminal EGFP, and encode 722 residues. The two single mutants and their combination allow site-specific and combined comparisons; they are not three interchangeable "constitutively nuclear" positive controls [2, 12].

Table 5: Exact designed codon changes; plasmid coordinates are 1-based inclusive.

Variant	Codon change	Plasmid nucleotides	Comparison enabled
S142A	AGC → GCC	1054–1056	Recovery with or without the S142 phospho-acceptor
S211A	TCC → GCC	1261–1263	S211-dependent distribution and 14-3-3 association
S142A/S211A	Both changes	Both intervals	Combined perturbation versus each single mutant

Generate each mutant using a sequence-defined synthetic donor with the same cloning flanks as the accepted WT amplicon, followed by the established HindIII/BamHI workflow. The supplied donor FASTAs specify the intended DNA; they are not purchased or experimentally validated material. Existing cloning primers may amplify the matched donors, but new mutagenesis primers are not invented. Include no-template PCR, digested-vector-only ligation and the established positive transformation control for each batch. Sequence the full fusion region and both junctions in both directions, explicitly including both target codons. Use two independently recovered, sequence-correct bacterial clones per genotype for key-contrast confirmation to detect clone/preparation-specific effects; randomize clone identity across experimental days. Bacterial clones are not additional biological cell-culture replicates.

The derived sequences retain the WT reading frame and unique HindIII/BamHI cleavage geometry. Ordinary diagnostic band sizes therefore do not identify these point mutants; the existing gel predictions apply to their gross insert/backbone sizes only. Sequence acceptance, intact expression and cell phenotype are three distinct checks. Alanine removes the serine phospho-acceptor but can also alter local interactions; none of these substitutions alone proves the responsible kinase or a direct phosphorylation reaction. A phosphomimetic S-to-D/E construct is not included as a guaranteed phosphorylated-state control.

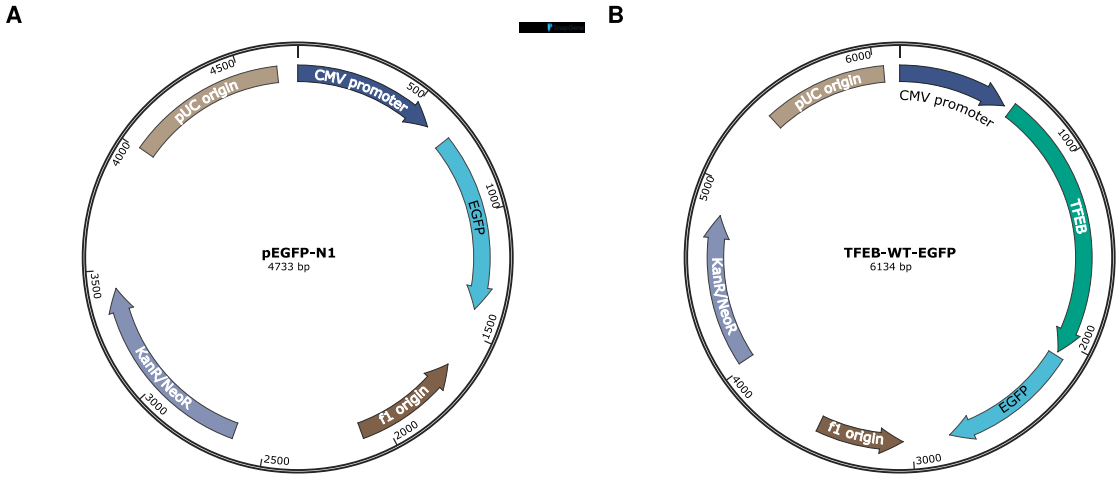


Figure 1: SnapGene Viewer maps of (A) the 4,733-bp EGFP-only control and (B) the 6,134-bp WT TFEB–EGFP reporter. Selected features retain their sequence coordinates and use a consistent NPG palette. The linker and minor regulatory annotations are omitted at this scale; complete GenBank records accompany the design. Native SVG geometry and vendor credit are preserved; the size unit is typeset in English. These are designed constructs, not verified experimental clones.

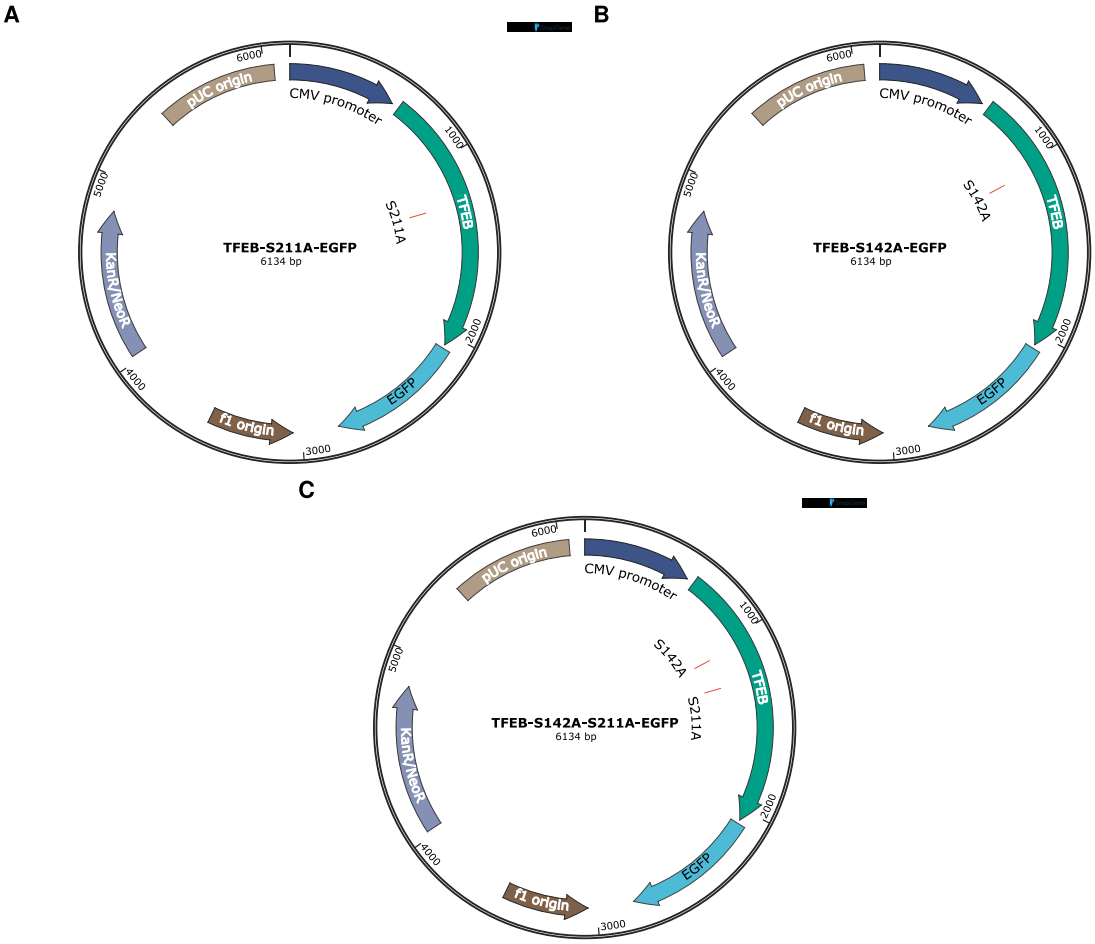


Figure 2: SnapGene Viewer maps of (A) S211A, (B) S142A and (C) S142A/S211A. Red marks identify the substituted codons; residue numbering refers to NP_009093.1. Every plasmid remains 6,134 bp and encodes 722 residues. EGFP, backbone and linker sequences remain identical to WT. These substitutions define experimental comparisons and do not establish a constitutively nuclear phenotype.

2.9 Questions and Control Structure

The expanded study tests reproducibility, reversibility, endogenous consistency, phosphorylation-site contributions, and TFEB-dependent function. Table 6 links each alternative explanation to a comparison. Modules use separate matched plates, rather than crossing every factor in one experiment. Numerical cell-assay settings below are proposed starting conditions or prespecified design choices, not locally optimized measurements.

Table 6: Control matrix: comparison, readout and interpretation.

Question	Required comparison	Readout and decision
Background / reagent	Parental; reagent-only mock; EGFP-only; WT fusion, under corresponding treatments	Background, morphology and viability. Mock tests reagent effects; EGFP-only tests tag/vector effects, not endogenous function.
Nutrient response	Amino-acid-replete vs depleted medium, with matched glucose, salts and dialyzed serum; separate serum-withdrawal arm	TFEB N/C and pathway markers. Do not equate serum withdrawal with amino-acid deprivation.
Drug response	Torin1 vs equal DMSO; untreated vs DMSO	N/C, p-S6K/total S6K, viability. Solvent alone must not reproduce the response.
Reversibility	Starve 60 min, then refeed 15/30/60 min; continued starvation and continuous feeding at equal total times	Change from starvation-endpoint log N/C; recovery of signaling. Controls for elapsed time and continued deprivation.
Tag / abundance	Endogenous TFEB in parental cells; reporter at two acceptable expression doses	Endogenous IF/fractionation; reporter expression-stratified N/C. Disagreement limits reporter interpretation.
Phosphorylation sites	WT, S211A, S142A and S142A/S211A, same backbone and treatments	Baseline, treatment interaction and recovery. Do not assume any mutant is always fully nuclear.
mTOR attribution	Non-targeting siRNA; two independent RPTOR siRNAs; two independent RICTOR siRNAs	Target depletion, p-S6K, p-AKT(S473), total counterparts and TFEB N/C. Verify both pathway perturbations.
14-3-3 association	WT vs S211A, vehicle vs Torin1; EGFP-only; beads-only	GFP pull-down, input and recovered bait; normalize bound 14-3-3 to precipitated fusion. Tests association, not direct contact.
TFEB dependence	Two individual TFEB 3′-UTR siRNAs; non-targeting control; empty vector; coding-only WT rescue; control-siRNA + WT	Endogenous depletion, rescue abundance and target transcription. Reproducible loss plus rescue supports a TFEB contribution.
Autophagic turnover	Each stimulus and knockdown/rescue condition with and without terminal bafilomycin A1	Inhibitor-dependent LC3-II increment, loading and viability. Raw LC3-II abundance is not flux.
Assay specificity	No-primary IF; phosphatase-treated lysate; S211A; fraction-purity markers; qPCR no-template/no-RT	Reject nonspecific staining, phospho-signal, mixed fractions or contamination before interpreting biology.

2.10 Cell Quality, Expression and Treatment

Use authenticated, mycoplasma-negative HeLa cells from a recorded source and restricted passage range. Distribute the same cell suspension across each experimental block. Culture in DMEM with 10% FBS at 37 °C and 5% CO_2, aiming for 60–70% confluence at transfection. Pilot 0.10, 0.25 and 0.50 μg fusion plasmid per 35-mm dish with one lipid reagent at its documented DNA ratio; evaluate expression after 24 h. The range is a project choice informed by published HeLa transfection work [33]. Select the lowest dose giving adequate nonsaturated signal and retain a second acceptable dose for sensitivity analysis. Match plasmid molarity across constructs and total DNA mass using a verified nonexpressing carrier. Equal mass alone does not match molecule number for the 4,733-bp empty and 6,134-bp fusion vectors.

The primary nutrient comparison uses a defined DMEM-based formulation with or without its amino-acid supplement, including glutamine; glucose, salts, vitamins and 10% dialyzed serum remain matched. Refeeding restores that amino-acid mixture after 60 min. A separate amino-acid-replete, serum-free arm tests serum withdrawal. Freeze the medium formulation and supplier record before execution. Fed and depleted wells receive matched washes and medium exchanges.

The proposed Torin1 starting condition is 250 nM for 60 min, preceded by a 125/250/500 nM pilot at 30/60 min. Literature establishes responsiveness, not optimization of this transient-expression protocol [2]. Match DMSO to the highest concentration used, capped at 0.1%, in relevant controls. Select the lowest exposure with reproducible pathway engagement and acceptable viability. Prespecify at least 90% membrane-intact cells and no more than a 10-percentage-point loss versus vehicle, including detached cells. Reject unsuitable conditions during the pilot, before inspecting confirmatory TFEB effects.

Collect acute localization at 0/15/30/60 min in separate matched wells. For recovery, collect after 60-min starvation and after 15/30/60-min refeeding; include continuously fed and continued-starvation wells at 75/90/120 min total time. Apply this series to WT and all three mutants. RNA, co-IP and flux use separate plates. Figure 3 shows acute and recovery sampling.

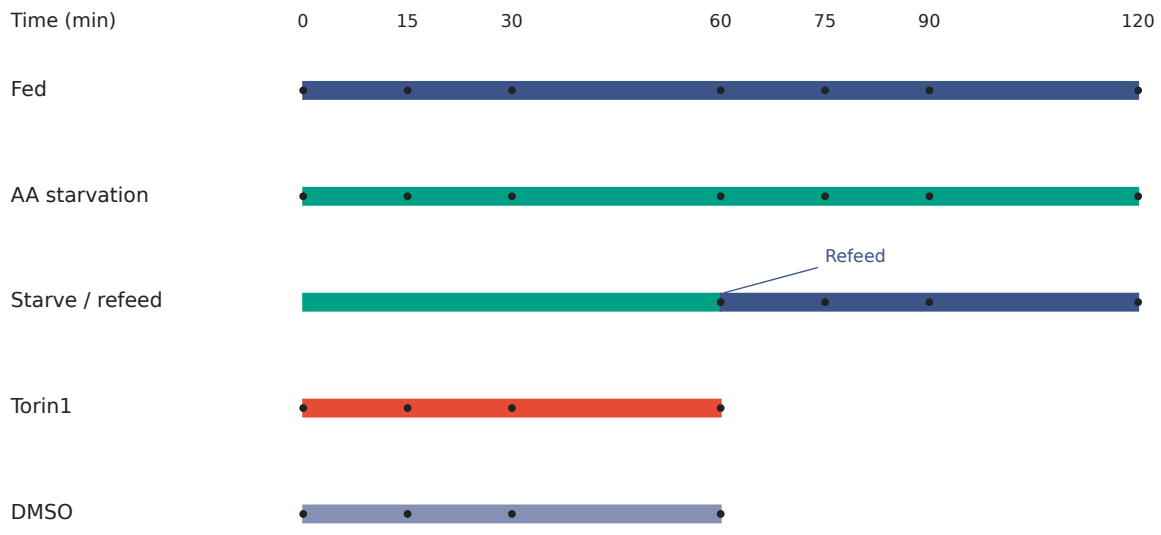


Figure 3: Planned sampling schedule, not response curves. Recovery has both continuous-feeding and continued-starvation controls at matched total times. Every point represents separate matched wells. AA denotes amino acids; blue indicates amino-acid-replete medium, green depleted medium, red Torin1 and purple matched DMSO. WT and all three mutants share the recovery series; later transcription and flux assays use separate plates.

2.11 Imaging, Endogenous Confirmation and Sampling

Start with 4% paraformaldehyde fixation for 10 min and DAPI counterstaining; check GFP preservation in the pilot. Optimize permeabilization and endogenous TFEB antibody dilution in parental cells. Include no-primary staining and show signal reduction with each TFEB siRNA before quantitative use. Keep acquisition settings identical and nonsaturating within each day; randomize plate positions and acquisition order, code images for analysis and retain raw files.

Segment nuclei with DAPI and cells with a compatible boundary marker or validated transmitted-light method, rather than a treatment-dependent GFP threshold. Exclude nuclei and neighboring cells from cytoplasmic masks. Use background-corrected mean intensities:

$$R_{\mathrm{N/C}} = \frac{I_{\mathrm{nucleus}} - I_{\mathrm{background}}}{I_{\mathrm{cytoplasm}} - I_{\mathrm{background}}}. \tag{1}$$

Retain component intensities. Exclude out-of-focus, saturated, mitotic, dead, overlapping or unsegmentable cells under pilot-frozen rules. Cytoplasmic intensity at or below background plus three background standard deviations is nonquantifiable; report its frequency rather than assigning an arbitrary large ratio. Report exclusions by group to detect treatment-dependent selection.

Plan four independent culture/transfection days after a separate pilot, two technical wells per condition, and at least 30 eligible cells per well from at least five prespecified nonoverlapping fields. Analyze all eligible cells in those fields or randomly subsample with a recorded seed. Cells, fields and technical wells are nested observations, not independent biological replicates. Confirm key WT treatment/recovery contrasts in parental cells by endogenous TFEB IF and fractionation. Agreement supports endogenous consistency, not equivalence between tagged and native protein.

2.12 Protein Expression, Phosphorylation and Fractionation

Use anti-GFP and anti-TFEB whole-cell blots to check intact expression. The WT fusion has 722 residues and a theoretical unmodified mass of 80.42 kDa; mutants retain its length but not its exact mass. A low band is not assigned as cleavage solely by size. Establish linear detection with a pilot loading series, then start with 15–20 μg protein per lane. Preserve phosphorylation with appropriate inhibitors, show uncropped membranes and markers, and normalize abundance to total-protein staining. Use a housekeeping band only after checking stability.

Probe p-TFEB(S211)/total TFEB at the fusion-sized band and p-S6K(T389)/total S6K. S211A and a separate phosphatase-treated aliquot test phospho-antibody specificity. Use compatible phosphatase buffer without inhibitor for that aliquot and matched handling for the untreated aliquot. Do not mix endogenous and fusion bands in the phospho/total ratio. p-S6K is pathway engagement, not a substitute for p-TFEB.

For nuclear/cytoplasmic fractionation, compare matched cell equivalents and correct for fraction volumes/recovery. Lamin B1 or histone H3 marks nuclei; tubulin marks cytoplasm. Assess reciprocal contamination and retain whole-cell input. Independently equal-mass-loaded fractions alone cannot establish redistribution. Require clean fractions and agreement with endogenous IF for a supported endogenous-localization claim.

2.13 Site Perturbations, Genetic Attribution and Binding

The sequence-defined panel is described in Section 2.8. S211A tests loss of that phospho-acceptor; S142A and the double mutant test additional recovery-associated regulation. S211A can retain Torin1 re-

sponsiveness [2]. Refeeding-related nuclear clearance measures net redistribution. S142/S138 have been linked to nuclear export [12], but fixed-cell N/C curves cannot separate import and export rates. A specific export-kinetics claim would require live-cell cytoplasmic FRAP with unbleached/photobleaching controls, or an export-inhibitor comparison.

For mTOR-complex attribution, test two nonoverlapping siRNAs per target individually against RPTOR or RICTOR, plus equal-concentration non-targeting siRNA and reagent-only mock. Start a pilot at 10 nM for 72 h, with reporter transfection during the last 24 h. Verify depletion in each experimental block and assay p-S6K/total S6K and p-AKT(S473)/total AKT. Compare each depletion with non-targeting control under fed conditions; retain Torin1 as a response reference. A negative RICTOR result is informative only with verified depletion and a measurable mTORC2 readout; include a matched stimulated positive-control lysate if basal p-AKT is too low. This genetic comparison is literature-supported [10]; Torin1 alone cannot establish mTORC1 specificity. RNAi-resistant RPTOR rescue would be required for a stronger target-specific RNAi causal claim; that additional construct is not sequence-finalized here.

For co-IP, compare WT/S211A under vehicle/Torin1 using GFP-affinity capture from equal total-protein inputs. Include EGFP-only, beads-only and input fractions; if antibody-based IP is used, include a matched isotype control. Detect 14-3-3 and recovered fusion; normalize co-precipitated 14-3-3 to recovered TFEB–EGFP. Reduced association with equal bait recovery supports phosphorylation-associated complex formation. Nonspecific bead binding or unequal bait recovery prevents that inference. Co-IP alone does not establish direct contact.

2.14 TFEB Depletion, Rescue and Target-Gene Expression

Select two independent siRNAs against the TFEB 3′ UTR, tested individually, so the coding-only B-v1 transgene lacks the intended sites. Before ordering, verify targets against NM_007162.3 and expressed isoforms, check off-target complementarity and record exact sequences. Resistance requires both absence of the target site and maintained rescue expression during RNAi; it is not claimed from the label "3′-UTR siRNA." No new siRNA is presented as sequence-validated here.

For each of fed vehicle and sustained Torin1 include: non-targeting siRNA + empty vector; each TFEB siRNA + empty vector; each TFEB siRNA + WT; and non-targeting siRNA + WT. Include reagent-only controls in the pilot. Match plasmid molarity/total DNA; quantify endogenous depletion separately from fusion expression and titrate rescue close to endogenous abundance. The control-siRNA + WT arm detects effects of rescue overexpression. Start with 72-h RNAi and WT rescue during the final 24 h. Mutant rescue is secondary to successful WT rescue.

Measure *MCOLN1*, *LAMP1* and *CTSD* transcripts at 4/8 h on separate plates with matched time controls. The panel and timing are project choices testing output of the established TFEB transcriptional program [35]. Pilot *RPLP0*, *HPRT1* and *TBP*, retaining at least two stable reference genes. Validate new qPCR primers for specificity, a single product, 90–110% efficiency, quantitative range and clean no-template/no-RT controls; document RNA quality and an inter-run calibrator [1]. Exact oligos remain a procurement-stage requirement. Analyze normalized Cq differences; use $2^{-\Delta\Delta Cq}$ only when efficiency assumptions hold. Technical PCR wells do not increase biological n. These mature-RNA measurements reflect both synthesis and decay; a transcription-rate claim requires the additional nascent-RNA controls specified in the research extension.

A response reduced by both TFEB siRNAs and restored by WT supports a TFEB contribution. Residual induction may reflect incomplete depletion, TFE3/MITF compensation or parallel regulation, rather than prove TFEB independence. If substantial residual induction persists, assay TFE3 abun-

dance/localization before claiming exclusivity. Failed rescue is interpretable only after confirming rescue abundance, localization and RNAi resistance.

2.15 Autophagic Turnover and Toxicity

On separate plates measure LC3-I/LC3-II and SQSTM1/p62 with/without a terminal 60-min pulse of 100 nM bafilomycin A1 during a proposed 4-h Torin1 exposure. First pilot 50/100 nM for 30/60 min for blockade and tolerability; these doses are not established optima here. Each stimulus and knockdown/rescue condition needs its own inhibitor-free and inhibitor-treated well: vehicle, bafilomycin alone, stimulus alone and combination. Define operational turnover as the within-condition inhibitor-dependent increment of normalized LC3-II. Compare increments across stimulus and TFEB loss/rescue, not raw band size. This paired-inhibitor logic follows experimental TFEB/autophagy work [35].

Measure p62 transcript alongside protein because transcription changes can confound protein abundance. Check viability and loading in every arm. More LC3-II without additional inhibitor-dependent accumulation is compatible with impaired clearance. mTOR inhibition also acts through TFEB-independent pathways; TFEB depletion need not abolish the entire response. Do not use bafilomycin-treated wells to infer unperturbed TFEB localization. If turnover remains ambiguous, add a separate tandem mCherry–GFP–LC3 assay with inhibitor and single-channel controls; this is a conditional orthogonal assay, not a sequence-designed plasmid in this panel.

2.16 Prespecified Analysis and Decision Rules

Primary contrasts are WT amino-acid starvation vs replete medium at 60 min; WT Torin1 vs vehicle at 60 min; and WT 30-min recovery vs continued starvation at 90 min total time. Average cell $\log_2$(N/C) within well and then independent day/condition. Analyze paired day-level contrasts, report effects and 95% confidence intervals, and apply Holm correction across the three primary tests. Four days is a minimum replication plan, not a power calculation: use separate pilot variance and a stated meaningful effect to determine confirmatory sample size before collecting that dataset.

For mutants use a genotype-by-condition/time model with day as a block and prespecified WT-versus-mutant interaction contrasts. Account for each genotype's starvation-endpoint ratio in recovery analysis; different initial ratios are not automatically different recovery rates. If sample size/model assumptions fail, report day-level contrasts with uncertainty and collect the planned extra replicates. RNAi/rescue, transcripts, co-IP and turnover are separate secondary families with multiplicity control. Plot independent-day summaries with nested cell distributions. A nonsignificant result is not equivalence. Discordant assays remain visible and trigger the relevant control checks; no effect sizes, error bars or significance are invented for this proposal.

2.17 Execution Plan and Deliverables

The core program is organized into twelve modules with explicit outputs (Table 7). The timeline is a proposed eight-week allocation after reagent availability, not a record of work performed. Five additional research modules are specified separately in the next section. Modules advance only after their stated acceptance checks. Figure 4 connects the different levels of evidence; arrows indicate the planned order of investigation, not demonstrated causation.

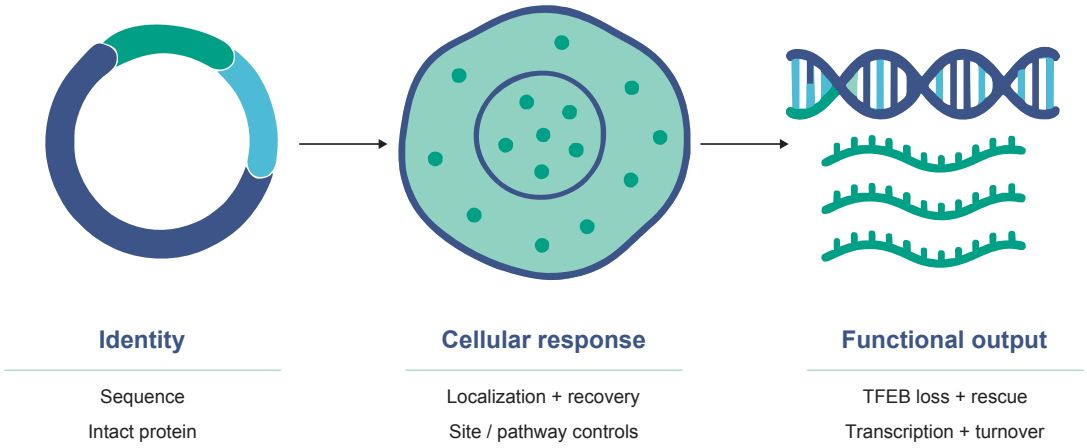


Figure 4: Evidence structure for the expanded study. Construct identity precedes biological assays; reporter and endogenous localization are checked before site/pathway interpretation. TFEB loss/rescue tests target-gene RNA responses and turnover. The transcription branch requires qualified nascent-RNA evidence for a rate interpretation; turnover requires paired bafilomycin A1 controls. Icons are conceptual drawings, not experimental images. Each stage needs the negative, positive and specificity controls in the methods. Agreement strengthens interpretation; disagreement narrows the conclusion.

Table 7: Proposed workload, acceptance checks and assessable deliverables.

ID	Week	Module	Deliverable / acceptance check
M01	1	WT and three mutant donor/construct designs	Versioned sequences, five plasmid maps, codon and junction audit. Computational deliverables are provided; material identity still needs verification.
M02	1–2	Cloning, selection and clone QC	PCR/digest records, failed-clone log and full bidirectional sequence acceptance; two independent accepted clones per genotype for key checks.
M03	2–3	Cell quality and expression pilot	Authentication/mycoplasma record; dose, viability and detector-linearity decisions frozen before confirmation.
M04	3–4	WT nutrient and drug response	Matched-media/solvent controls; raw images, masks, per-cell intensities and independent-day N/C summaries.
M05	4	Endogenous TFEB and fractionation	Specific endogenous IF; reciprocal fraction-purity checks and volume-corrected partition estimates.
M06	4–5	Time course, refeeding and mutants	Four-genotype baseline/treatment and recovery contrasts; matched elapsed-time controls and clone-confirmation record.
M07	5	Phosphorylation and pathway readouts	Uncropped p-TFEB, p-S6K and total-protein blots; S211A and phosphatase specificity checks.

ID	Week	Module	Deliverable / acceptance check
M08	5–6	RPTOR versus RICTOR RNAi	Two individual siRNAs per target, knockdown and pathway-engagement evidence; qualified complex-attribution conclusion.
M09	6	14-3-3 association	GFP capture with input, EGFP-only and beads-only controls; association normalized to recovered bait.
M10	6–7	TFEB loss and WT rescue	Two independent 3′-UTR siRNAs; endogenous depletion, verified rescue expression and control-siRNA + WT arm.
M11	7–8	Transcription and turnover	Validated qPCR assays and paired inhibitor/no-inhibitor LC3-II contrasts; p62 transcript/protein and viability.
M12	8	Integration and reproducibility	Blinded analysis record, exclusions, effect estimates with intervals, source data, figure legends and alternative-interpretation table.

For a concrete workload example, the four fusion genotypes crossed with three matched 60-min conditions (fed vehicle, amino-acid deprivation with vehicle, and fed Torin1) yield 12 conditions. Four independent days with two technical wells per condition require 96 assay wells and at least 2,880 eligible cells at 30 cells per well. These are planning quantities, not observations; they exclude parental/mock/EGFP-only controls, pilot wells, recovery time points, protein/RNA plates and repeat attempts. A separate plate map must enumerate those additions before execution. Independent-day replication must not be replaced by more cells from one day. The larger program is divided into modules to keep each comparison interpretable and scheduling feasible.

The report currently supplies design figures and computational sequence checks. Experimental image, blot, qPCR and turnover panels are planned outputs and will only be populated with acquired data. Before running M08–M11, finalize exact siRNA targets, qPCR oligos and reagent identities and verify their performance; these are open execution items rather than completed designs.

3 Hypothesis-Driven Research Extension

3.1 From Replication to a Discriminating Question

The localization experiments above reproduce established TFEB biology. The extension asks a different question: *does the effect of the C-terminal S466/S467/S469 cluster on target-gene RNA abundance depend on the stimulus and fusion architecture?* Protein abundance and nuclear exposure are measured as possible mediators. An architecture effect is a hypothesis to test, not an explanation already established by the literature.

Paquette et al. reported that the S466A/S467A/S469A mutant (3A) could accumulate in nuclei yet showed impaired transcriptional activity in their experimental systems; they also examined combinations with S142A/S211A [30]. Negoita et al. subsequently observed AMPK-associated dephosphorylation of this cluster and explicitly raised the possibility that alanine substitutions affect local structure as well as phosphorylation [14]. Their study also distinguished AMPK-dependent responses to MK-8722 from

responses to Torin1. Different cell systems, stimuli and constructs prevent treating these findings as a single matched comparison. Neither 3A nor the five-site alanine combination is a new mutant invented here.

Our proposed contribution is a controlled genotype–architecture–stimulus comparison in HeLa cells, followed by selected mechanistic tests. Negoita et al. also obtained supporting phosphosite evidence from C-terminal GFP knock-in material [14]; the literature disagreement cannot simply be assigned to GFP. The 3T substitution retains hydroxyl-bearing side chains but remains potentially phosphorylatable; it is a sequence/chemistry probe, not a constitutively dephosphorylated mimic. A targeted literature search supports this rationale but does not establish that the exact combination has never been tested. No claim of first discovery is made.

The two architectures differ in linker, EGFP coding sequence, protein length and the position of the stop junction. Thus "tag" in the matrix is shorthand for the entire GDPPVAT–EGFP fusion architecture. A difference cannot identify EGFP itself as the causal component. Attribution to EGFP alone would require sequence-accepted linker-only and appropriately matched fusion controls in a subsequent design; these are not among the five finalized constructs. Drug-associated changes in transgene RNA or protein stability must also be checked before proposing a protein-level tag mechanism.

3.2 Five Additional Sequence-Defined Constructs

The existing WT fusion is reused. Five additional plasmid designs provide the remaining cells of the six-construct matrix (Table 8). All retain the same CMV promoter and bacterial backbone. Full GenBank, DNA FASTA and translated protein files are supplied in `research-v3`; circular maps are shown in Figures 5 and 6.

Table 8: Research constructs and exact intended changes. Coordinates refer to the accepted 6,134-bp WT fusion.

Construct	bp / aa	Intended difference from the WT fusion
3A–EGFP	6134 / 722	S466A/S467A/S469A; each AGC codon becomes GCC.
3T–EGFP	6134 / 722	S466T/S467T/S469T; each AGC codon becomes ACC.
WT, untagged	5396 / 476	Replace bp 2059–2799 (linker plus EGFP including stop) with the native TFEB stop codon TGA.
3A, untagged	5396 / 476	The same tag removal plus the three AGC-to-GCC changes.
3T, untagged	5396 / 476	The same tag removal plus the three AGC-to-ACC changes.

The three codons occupy bp 2026–2028, 2029–2031 and 2035–2037; intervening F468 is unchanged. Tag removal shortens the plasmid by 738 bp and eliminates the original BamHI site. Therefore the WT HindIII/BamHI diagnostic and its expected fragment sizes cannot be reused for untagged constructs. For this extension, the proposed procurement route is sequence-defined whole-plasmid synthesis followed by independent whole-plasmid sequence verification, including every altered codon, the new stop junction and the backbone. No order has been placed and no assembly primers have been experimentally validated. Two independently recovered, sequence-accepted bacterial clones per construct will confirm key comparisons; they are not counted as independent culture-day replicates.

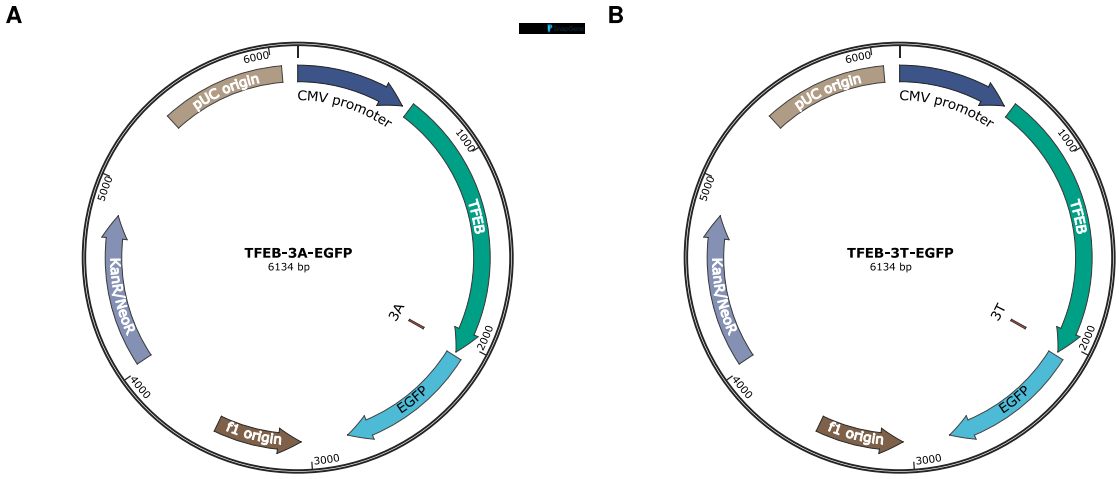


Figure 5: Additional tagged constructs rendered by SnapGene Viewer. (A) TFEB-3A–EGFP; (B) TFEB-3T–EGFP. Both are 6,134 bp. The red feature marks the S466/S467/S469 cluster; its span includes unchanged intervening sequence. CMV is blue, TFEB teal, EGFP cyan, selection purple and replication origins brown. Maps show sequence designs, not experimentally recovered plasmids.

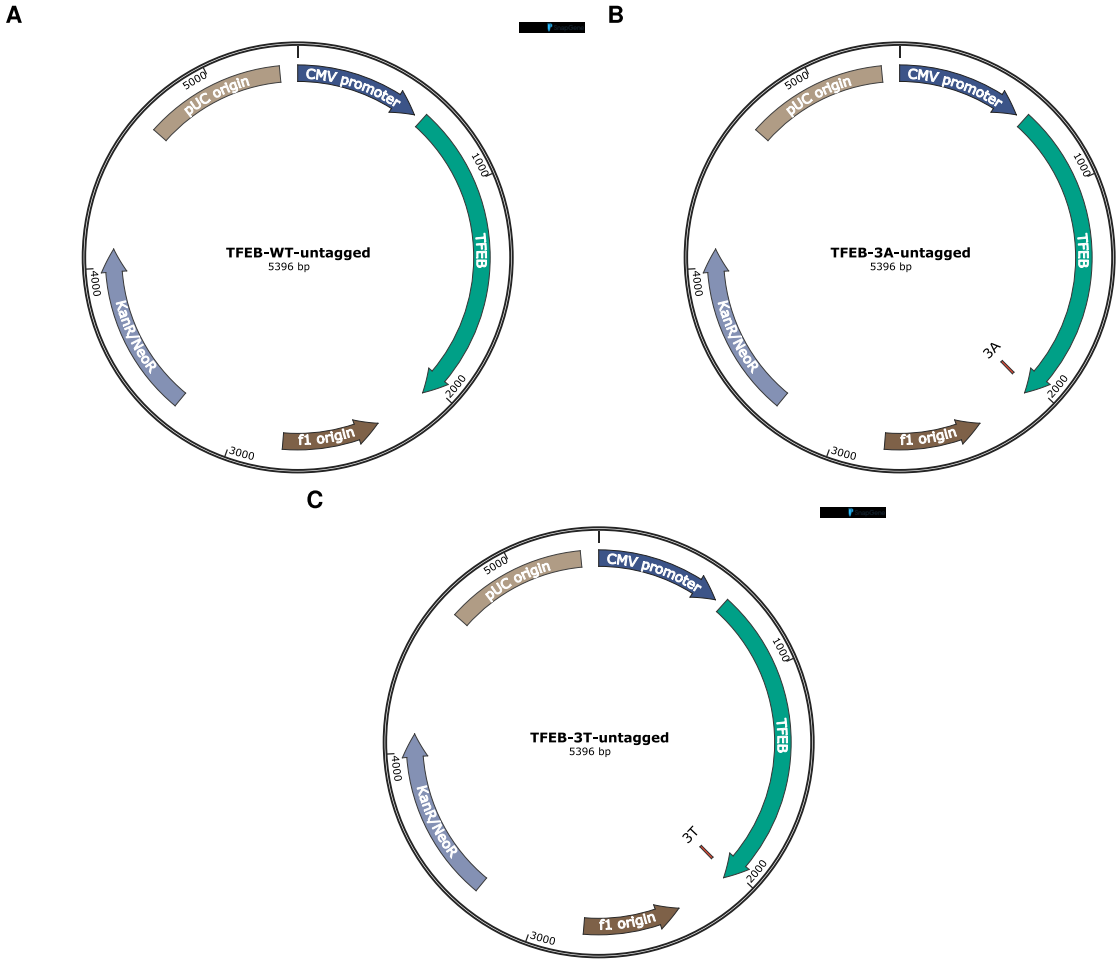


Figure 6: Matched untagged constructs rendered by SnapGene Viewer: (A) WT, (B) 3A and (C) 3T. Each is 5,396 bp and encodes a 476-residue TFEB protein followed by its native stop codon. Removing the linker and EGFP preserves the remainder of the parental backbone while changing downstream coordinates and the restriction pattern. Feature colors match the tagged maps. Full annotations and independent sequence checks accompany the design files.

3.3 Core Factorial Comparison and Controls

Cross WT/3A/3T with EGFP-tagged/untagged TFEB and vehicle/Torin1/MK-8722, yielding 18 core conditions (Figure 7). Perform the principal comparison after endogenous TFEB depletion with one validated 3′-UTR siRNA absent from the coding-only constructs. Repeat the complete prespecified WT/3A comparison (both architectures and all three treatments), not just its largest effect, with a second individual TFEB siRNA in separate culture-day blocks. Verify transcript coverage across expressed TFEB isoforms and maintained transgene expression during RNAi. Quantify endogenous depletion by 3′-UTR RT-qPCR and protein in matched no-rescue wells. Untagged rescue and residual endogenous TFEB cannot be separated by size; no-rescue wells bound this contamination but do not measure its exact amount in each rescued cell.

Require acceptable viability, at least 80% endogenous protein depletion in no-rescue wells and pretreatment median TFEB signal per expressing cell within 0.5–2 times the parental reference. These numerical gates are project design choices, not published assay-performance measurements. Untagged WT must restore at least half of the pilot-estimated loss of drug-induced MCOLN1 response, in the same direction, in each drug context for which a TFEB-dependent interpretation is sought. The loss must exceed the pilot assay noise. If this cannot be demonstrated, the drug comparison remains descriptive until the assay is repaired. A defective tagged WT is itself a possible architecture result and is not discarded because it fails untagged-WT rescue performance. Acceptance gates are frozen before confirmatory genotype outcomes are unblinded.

The following controls are run at vehicle, Torin1 and MK-8722 in each independent block. A machine-readable condition manifest accompanies the proposal; counts include culture conditions, not qPCR technical reactions.

Table 9: Research condition inventory. NT is non-targeting siRNA; KD is the selected individual TFEB siRNA. A validated nonexpressing DNA carrier supplies the no-rescue DNA control.

Arm, crossed with three treatments	Primary	Second-siRNA confirmation
KD + WT/3A/3T, both architectures	18	12 (WT/3A)
NT + WT/3A, both architectures	12	12
NT or KD + nonexpressing carrier, no rescue	6	6
NT or KD + EGFP-only	6	6
Reagent-only mock, neither nucleic acid	3	3
Total conditions per independent day	45	39

NT + WT/3A arms test whether a genotype effect depends on the depletion background; no-rescue arms test loss and baseline drug responsiveness. EGFP-only does not substitute for a nonexpressing control. Add an untreated/no-solvent comparison in the pilot. Keep DNA carrier identity, total DNA mass and total siRNA concentration constant within the DNA-transfected comparisons; reagent-only mocks intentionally lack nucleic acid. DNA molarity is a starting calibration, followed by expression titration rather than a simultaneous claim of equal dose and equal expression. Randomize positions and processing order, balance every contrast across plates, and include shared reference samples if a complete day block spans plates. Record plasmid lot, endotoxin/purity QC and clone identity; do not confound one genotype with one preparation batch. Confirm key effects with the second sequence-accepted clone.

Pilot DNA amounts separately to overlap TFEB abundance and the fraction of expressing cells across constructs, using matched total DNA and transfection reagent. Quantify both architectures with the same anti-TFEB reagent recognizing an epitope outside the changed C-terminal region. Validate specificity by depletion and linearity with antigen-loading series for both architectures. Freeze each construct-specific DNA dose before drug exposure and use that dose in all three treatment arms; do not retitrate after treatment to erase a biological abundance response. In the pretreatment expression pilot, target a between-architecture ratio of 0.8–1.25 for the median TFEB signal among expressing cells and an expressing-fraction difference no greater than 10 percentage points. These are prespecified practical matching tolerances; report the measured differences and uncertainty, not a claim of equivalence based on nonsignificance. Freeze a common threshold above depleted/no-rescue background, use randomly selected fields, and report all-cell distributions, stratified summaries and exclusions. Do not use GFP brightness to select one architecture. If common expression support is absent, revise delivery before the confirmatory comparison. At baseline and 6 h, assay transgene RNA with validated backbone–TFEB-junction and shared TFEB-CDS assays, endogenous RNA with the $3'$-UTR assay, and protein separately; tag-dependent RNA abundance is an alternative to altered protein activity.

For the extension, use a common nutrient-replete medium and matched final solvent. Pilot MK-8722 at 1, 3 and 10 μM; retain the Torin1 125/250/500-nM pilot range. Select the lowest tolerated dose with reproducible intended engagement. These are proposed HeLa starting points, not validated operating conditions. Measure p-ACC/total ACC and p-S6K/total S6K under both drugs, plus p-TFEB(S211), TFEB abundance, cell counts and viability. Recheck engagement and tolerability at 6 h; an acute pilot does not qualify a longer exposure. Do not force the two drugs to have equal downstream effects or describe their comparison as a selective AMPK-versus-mTOR mechanism. Report their dose/kinetic differences and retain a second acceptable dose as a sensitivity experiment.

Monitor localization at 0, 30, 60, 120, 240 and 360 min in matched destructive samples, and collect the primary RNA endpoint at 6 h. Retain N/C, nuclear concentration, integrated nuclear TFEB signal and cell/nuclear size. N/C alone can be equal when nuclear amounts differ. Summarize the sampled nuclear-signal trajectory and trapezoidal area as a population exposure proxy; it is not a continuous single-cell trajectory or an absolute molecule count. An intrinsic transactivation interpretation requires overlapping nuclear-signal distributions and comparable trajectories, supported by orthogonal fractionation. Prespecify matching limits from assay precision before confirmation. Differences in exposure remain a potential biological mediator, not a reason to delete the affected samples or automatically adjust the primary effect away.

3.4 Primary Estimand, Replication and RNA Interpretation

The primary endpoint is normalized steady-state MCOLN1 RNA abundance at 6 h; LAMP1 and CTSD are secondary. Validate RNA integrity, genomic-DNA removal, primer efficiency, quantification limits and at least two reference genes across all treatments, architectures and depletion backgrounds, following qPCR reporting principles [1]. Include no-template/no-RT controls and an inter-run calibrator. A missing amplification is not zero expression; values outside the validated quantitative range are flagged, with a prespecified repeat/exclusion rule. Do not replace an unresponsive primary gene with whichever secondary gene is significant.

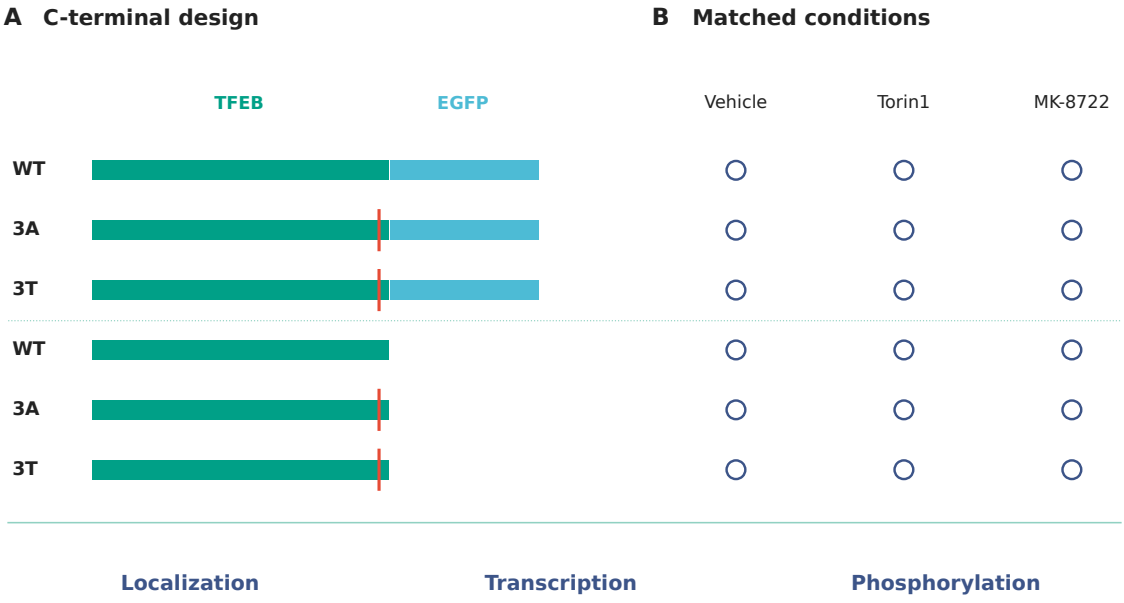


Figure 7: Factorial design for the research extension. (A) WT, 3A and 3T TFEB carry the C-terminal linker–EGFP architecture (upper rows) or lack the linker/tag (lower rows). Red ticks mark the altered cluster; small sequence features are omitted. (B) Three treatments give 18 core conditions; open circles are planned conditions, not data or replicates. The "Transcription" branch includes mature-RNA screening and conditional nascent-RNA confirmation; qPCR alone measures RNA abundance. Table 9 adds the full depletion/rescue, background and mock controls. Expression, pathway and assay checks are specified in the text.

Let $Y_{g,a,s,d}$ denote the mean $\log_2$ normalized RNA value from culture wells on independent day d, for genotype g, architecture a (E, EGFP fusion; U, untagged) and stimulus s (V, vehicle; M, MK-8722; T, Torin1). Average qPCR technical replicates within each sample first. Define

$$\Delta_{a,s,d} = (Y_{3A,a,s,d} - Y_{3A,a,V,d}) - (Y_{WT,a,s,d} - Y_{WT,a,V,d}), \tag{2}$$

$$C_d = (\Delta_{E,M,d} - \Delta_{U,M,d}) - (\Delta_{E,T,d} - \Delta_{U,T,d}). \tag{3}$$

The primary estimand is the mean C_d, the difference between the architecture-dependent 3A effects under the two drugs. Vehicle terms cancel algebraically from C_d, but remain essential to estimate each drug response and baseline genotype effects. Report those components even if the interaction is null: equal nonzero effects under both drugs can yield $C_d = 0$. The primary result is a total effect in the calibrated expression system. Post-treatment nuclear exposure and protein abundance are possible mediators and are not automatically entered as nuisance covariates; matched-exposure analyses are secondary and cannot alone establish a direct effect.

Use complete independent-day blocks with separate cell culture, RNAi and transfection preparations; distribute technical wells across plate positions. Report day-level C_d values, their mean and a two-sided 95% interval based on between-day variation (a t interval with $n-1$ degrees of freedom when its assumptions are defensible). Cells, wells, clones and PCR reactions do not inflate n. A failed well is not replaced by a cell-level replicate. Prespecify repeat rules from technical QC; if a required arm is unavailable, exclude that incomplete day from the paired primary estimate, retain the available data descriptively, record the reason and acquire a replacement complete block. Outcome size or direction is never an exclusion criterion.

Use a separate pilot of at least four complete day blocks to estimate the variance of C_d, then fix the confirmatory sample size before collecting confirmation data. Plan at least six independent days and increase that number as needed for a target 95% interval half-width of 0.5 $\log_2$ units, using a conservative variance estimate from the pilot. This precision target is a design choice, not an observed effect or an equivalence margin; retain the calculation and do not stop when significance is reached. If resources cannot meet the target, label the extension exploratory. Keep pilot and confirmatory data separate. The secondary family comprises the four MCOLN1 $\Delta_{a,s}$ contrasts and the primary-form C contrasts for LAMP1 and CTSD (six tests, Holm adjustment). 3T and non-targeting-background comparisons are explicitly exploratory. The second-siRNA confirmation re-estimates the same primary contrast with its own complete blocks and interval; do not pool the two siRNAs as independent replicates or select the favorable reagent.

Synthesis versus stability. Mature RNA integrates production and degradation [32]. Before calling a difference a transcriptional defect, repeat the prespecified WT/3A contrasts with a short, pilot-qualified 4-thiouridine pulse ending at 6 h and quantify newly synthesized MCOLN1 and a secondary target. Include matched unlabeled/background-purification controls, input RNA, labeled/unlabeled recovery standards, cell-count normalization, and checks for labeling toxicity and treatment-dependent incorporation. A recovery spike does not control cellular nucleoside uptake. Freeze pulse duration and performance criteria before confirmation; if incorporation differs, the assay cannot establish differential synthesis. Concordant nascent and mature RNA strengthens a production interpretation; discordance triggers RNA-stability/processing studies. Intronic RT-qPCR can be an exploratory precursor-RNA assay, but is not a definitive substitute. Without a qualified nascent-RNA assay, retain the conclusion "target-gene RNA abundance" rather than "transcription rate."

MiT/TFE compensation. TFE3 can induce MCOLN1 and related targets even when TFEB is depleted [11]. Measure TFE3 RNA, protein and localization in the prespecified WT/3A comparison and no-rescue controls, not only after an inconvenient result. If TFE3 activity or residual induction changes, test the key contrasts with TFEB-only, TFE3-only and combined depletion, matched total siRNA, two individual TFE3 reagents and WT-TFEB rescue. Recheck viability and rescue expression in each background. Persistence under competent combined depletion limits a TFEB/TFE3-specific interpretation; loss under TFE3 depletion indicates family-context dependence, not necessarily direct TFEB–TFE3 binding. MITF or other routes remain alternatives. No TFEB-exclusive claim is made from single-factor rescue alone.

3.5 Mechanistic Follow-Up with Defined Decision Rules

AMPK dependence. Prespecify WT/3A, both architectures and all three treatments for R03; choosing only the largest R02 difference would be exploratory. Co-deplete PRKAA1 and PRKAA2 using two independent siRNA pairs in separate experiments. Include non-targeting, each single-target depletion, no-rescue and mock controls; match the total siRNA burden, including the concurrent TFEB siRNA. Verify each catalytic subunit, MK-8722-induced p-ACC, p-S6K, viability, nuclear exposure and rescue abundance at the relevant times. A selective loss of the MK-8722 response with a retained Torin1 response supports stimulus-dependent AMPK involvement; Torin1 is a comparator, not a guaranteed AMPK-independent positive control. If both responses fall, evaluate loss of cellular competence and altered baseline expression rather than selecting one literature model. A strong target-specific causal claim additionally requires a near-endogenous, RNAi-resistant AMPK rescue with a matched empty-vector arm; the required isoform/complex and rescue constructs are not sequence-finalized here. Without it, re-

port convergent RNAi evidence for involvement, not definitive on-target mediation. None of these tests identifies a direct TFEB kinase or phosphatase.

Phosphosite specificity. In WT tagged and untagged samples, measure C-terminal phosphorylation together with total TFEB at baseline and after treatment. Phospho-antibodies require phosphatase-treated material, unmodified/phosphorylated peptide controls and site-mutant controls; adjacent residues can affect epitope recognition. Loss of signal in 3A alone is not evidence of physiological dephosphorylation. If specificity is unresolved, use targeted LC–MS/MS with site-localization evidence, appropriate peptide standards and total-TFEB normalization; report ambiguous adjacent-site assignments as such. Enrich both architectures through a validated common TFEB epitope outside the altered region, document input/recovery and include IgG or bead controls. GFP-only capture would systematically exclude the untagged arm. Relative phosphopeptide abundance is not absolute site occupancy without suitable calibration. Analyze 3T with an assay capable of detecting threonine phosphorylation; a serine-phospho-antibody cannot establish its state. This follow-up is conditional on instrument access and assay development.

Promoter engagement. Enter R05 only after a qualified RNA-production difference and comparable nuclear exposure are established. Test TFEB occupancy near validated MCOLN1 and LAMP1 regulatory regions by ChIP–qPCR at matched collection times. Include input, matched IgG, TFEB-depleted/no-rescue material, a negative genomic region and a positive locus with demonstrable WT enrichment. Use a common TFEB antibody outside the changed region; validate equivalent recovery, chromatin fragmentation and crosslinking conditions across architectures. Start from matched cell equivalents and report percent input; percent input alone does not correct for different nuclear TFEB abundance or antibody accessibility. Use a validated spike-in approach if global recovery differs. Reduced occupancy would support altered engagement only after these checks. Comparable occupancy with different nascent RNA would motivate cofactor studies, without proving a particular cofactor mechanism. If these checks fail, report occupancy as unresolved.

Table 10: Interpretive boundaries of possible extension outcomes; all outcomes are hypothetical.

Observed pattern, if obtained	Supported interpretation and remaining alternative
3A defect only in EGFP-tagged TFEB	Supports fusion-architecture dependence; linker, RNA, antibody and delivery alternatives remain. It does not isolate EGFP itself.
3A defect in both tag states	A tag alone is insufficient to explain the effect; phosphorylation and local structural changes remain alternatives.
3T resembles WT while 3A differs	Consistent with a contribution of local side-chain chemistry; 3T phosphorylation must be measured before attributing the difference to hydroxyl retention alone.
Different effects under Torin1 and MK-8722	Supports stimulus dependence after verified pathway engagement; does not identify a direct kinase or phosphatase.
Mature RNA differs but nascent RNA does not	Investigate RNA stability/processing and assay sensitivity; do not claim a transcription-rate defect.
Nuclear exposure or abundance also differs	Retain the total effect, but do not assign it to intrinsic transactivation.
No clear interaction with wide intervals	Inconclusive precision; also inspect component effects because equal effects under both drugs can cancel.

3.6 Additional Workload and Feasibility

The extension retains five modules after M01–M12: R01, verify the five constructs and qualify expression/depletion; R02, run the factorial comparison, second-siRNA confirmation and conditional RNA-production/family-context checks; R03, test AMPK dependence; R04, resolve phosphosite specificity; and R05, test promoter occupancy only after its entry criteria are met. Weeks 9–12 are a provisional preparation/pilot window, not a promise to finish confirmation and all mechanistic assays. R04–R05 and the conditional R02/R03 additions depend on assay development, resources and preceding results. The eight-week core plan excludes this extension.

At the minimum six confirmatory days and two culture wells per condition, the 18 research conditions alone require 216 wells. Including controls, the primary block requires $45 \times 6 \times 2 = 540$ culture wells, and the second-siRNA WT/3A confirmation requires $39 \times 6 \times 2 = 468$: 1,008 wells *per assay and endpoint* across the two campaigns. These are lower-bound planning counts, not a powered final sample size. They exclude the independent pilot, six-point destructive imaging series, RNA/protein duplicates on separate plates, additional doses, clone checks, R03–R05 and technical PCR reactions. Scaling to multiwell plates requires a density/delivery pilot; the 35-mm-dish doses are not copied unchanged. Release a physical plate map with all controls before execution. If capacity is insufficient, complete R01/R02 first and label the remaining modules deferred; retain all failed and null records. This avoids presenting an unbudgeted matrix as an executable four-week experiment.

4 Expected Results and Criteria for Success

4.1 Computed Construct and Experimental Predictions

The in-silico pEGFP-N1-TFEB-Bv1 (6,134 bp; Figure 8) encodes a 722-residue TFEB–EGFP fusion: 476 TFEB residues, the seven-residue GDPPVAT junction, and 239 EGFP residues. The native TFEB stop is absent; EGFP supplies the final TAA. The calculated unmodified average mass is 80.42 kDa.

The reconstruction and an independent standard-library audit agreed on sequence, reading frame, junctions, terminal stop and fragment geometry. Corresponding checks rejected four software challenges: reversed TFEB, a one-base frameshift, restored native TFEB stop, and an internal HindIII site. These are not experimental clones or a measured success rate.

Cloning PCR predicts a 1,457 bp product. HindIII/BamHI digestion yields a 1,439 nt insert and 4,695 nt backbone by the cut-to-cut single-strand convention. Their pydna duplex spans are 1,443 and 4,699 bp, including four-base overhangs; gel sizes use the nominal single-strand lengths. Because the empty vector loses a 38 nt MCS segment, the final length is not the intact vector length plus the undigested amplicon.

Table 11 and Figure 10 summarize screening. Flanking PCR distinguishes insert-bearing candidates from empty vector. Interpret a negative orientation assay only with an amplification-positive control. If all orientation assays are negative in the first round, sequence flanking-positive candidates. Reverse TFEB is a hypothetical challenge, not an expected product of incompatible-end directional ligation.

Flanking PCR and HindIII/BamHI digestion cannot exclude reversed TFEB; the proposed gel cannot resolve three-base retained-stop differences. Acceptance requires the expected digest and actual, unambiguous, concordant bidirectional sequence across 602–2799 (2,198 nt), including both junctions, the complete fusion ORF and terminal stop. Ten primers predict usable coverage in both directions at every target base, not experimental coverage or whole-backbone verification. Resolve low-quality calls,

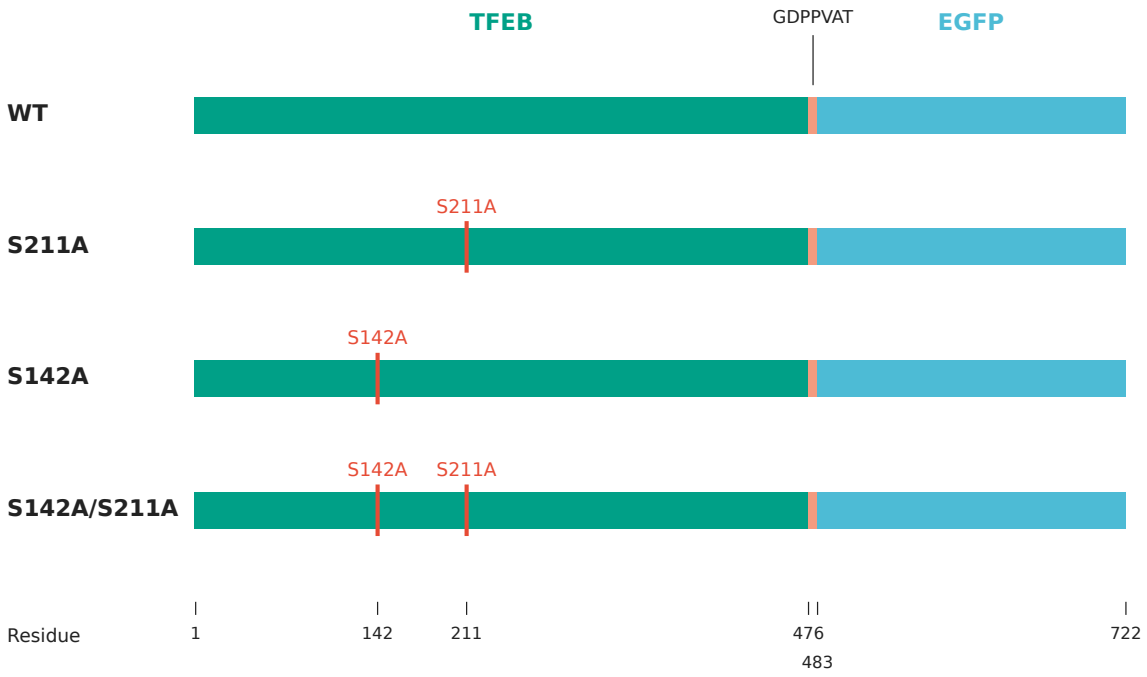


Figure 8: Aligned fusion-protein schematics for WT and three designed mutants. TFEB occupies residues 1–476, the GDPPVAT linker 477–483 and EGFP 484–722; rectangles are proportional to their encoded lengths. Red marks identify S-to-A substitutions. The accepted WT plasmid is pEGFP-N1-TFEB-Bv1 (6,134 bp). The full sequence retains HindIII and BamHI recognition starts at 622 and 2061; the first reference-strand bases after cleavage are 623 and 2062. Protein schematics do not show regulatory DNA or predict cellular activity.

Table 11: Predicted screening products. All sizes are nominal bp; "none" means no specific product is predicted. R and S are hypothetical challenge sequences.

Sample	Flanking PCR (S-F1/S-R1)	Orientation PCR (S-F2/S-R1)	HindIII/BamHI digest
Intended construct (C)	2,605	2,093	4,695 + 1,439
Empty vector (E)	1,204	none	4,695 + 38
Reverse TFEB (R)	2,605	none	4,695 + 1,439
Retained TFEB stop (S)	2,608	2,096	4,695 + 1,442

mixed peaks, gaps and unexpected target changes under the Sanger acceptance procedure before cell assays. Figure 9 summarizes the planned workflow and controls.

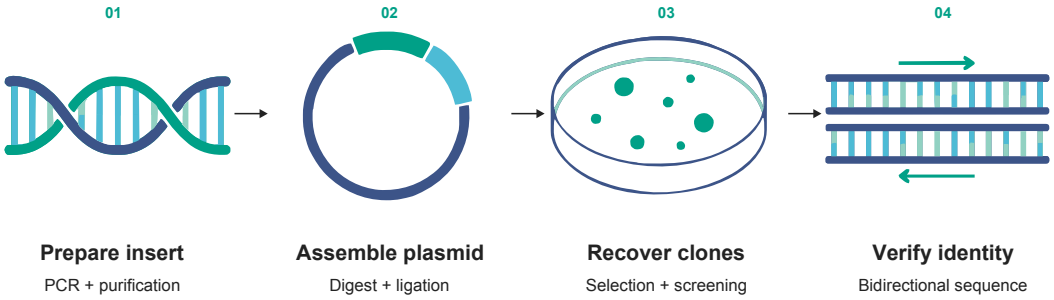


Figure 9: Proposed cloning workflow. Check ordered-template identity before PCR; include a no-template control, vector-only ligation, no-DNA transformation and a separate competence control. Purify backbone and insert after separate digests. Flanking and orientation PCR plus diagnostic digestion screen gross identity but cannot exclude point substitutions or a retained stop. Final acceptance requires unambiguous, concordant bidirectional sequence across the target and both junctions; failed candidates are resequenced or replaced. Measured DNA concentrations determine actual reaction inputs.

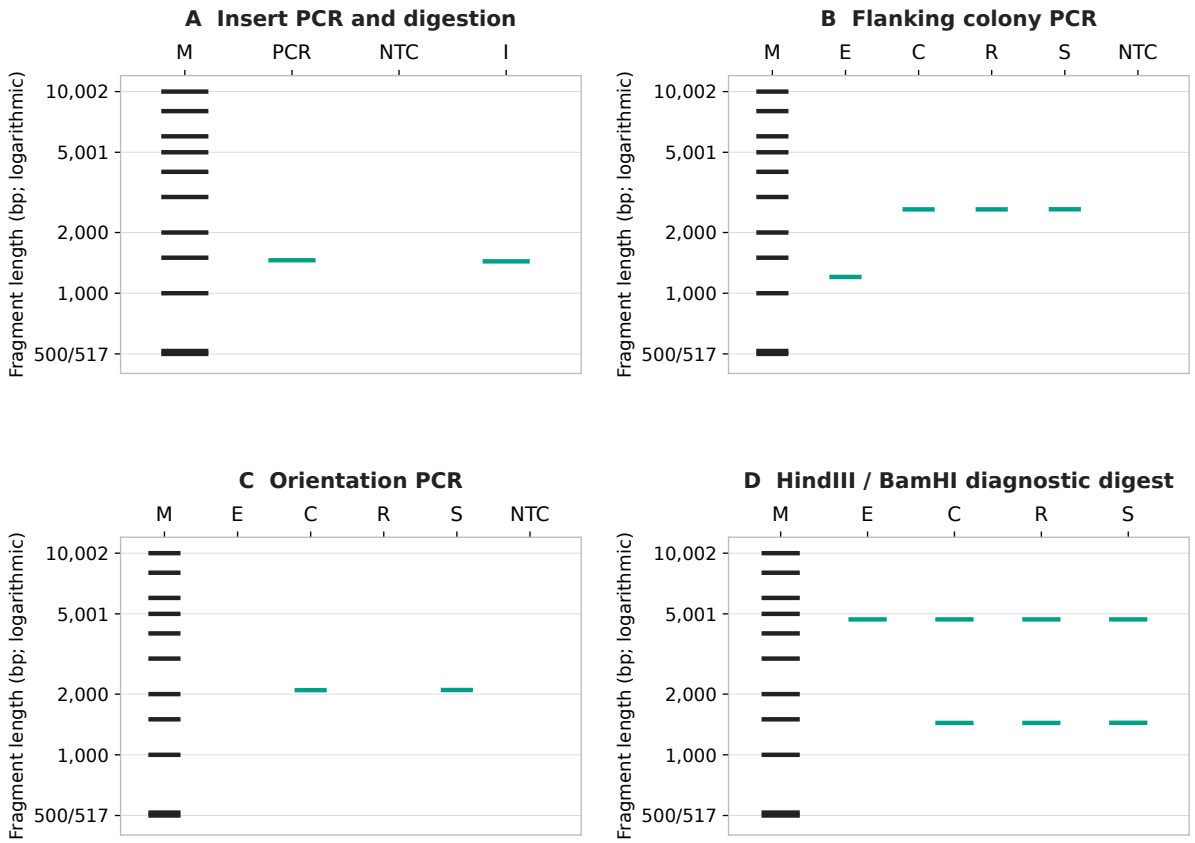


Figure 10: Predicted DNA-band positions, drawn as a schematic rather than an experimental gel. M: NEB N3232 ladder; PCR: tailed insert amplicon; I: digested insert; NTC: no-template control; E: empty vector; C: intended construct; R: hypothetical reverse-TFEB model; S: hypothetical retained-stop model. The 500/517 bp marker pair is not resolved [25]. Panels B and C use the flanking and orientation primer pairs defined in the screening table. The 7, 11 and 38 nt fragments fall below the displayed range. Three-base differences between C and S are not expected to resolve on the proposed 1% gel. An empty NTC schematic means no specific product is predicted, not a guarantee of a clean experimental lane. Uncut circular DNA mobility and band intensity are not predicted.

4.2 Localization, Mechanism and Function: Decision Matrix

The following are directional expectations and decision rules. No measured bands, microscopy fields, localization curves, effect sizes or statistical significance are supplied. A successful construct sequence alone is not a successful cell assay.

Table 12: How outcomes would support, limit or challenge the proposed explanation.

Test	Supporting pattern	Alternative result and next check
WT response	Higher N/C after amino-acid withdrawal or Torin1; lower N/C after refeeding than continued starvation	No change: inspect treatment engagement, viability, protein integrity and segmentation before rejecting responsiveness.
Endogenous agreement	Endogenous IF and clean fractionation show the same directional response as the reporter	Reporter-only response: test abundance dependence and label effects; restrict the finding to the reporter.
S211 contribution	S211A changes baseline distribution and reduces 14-3-3 association relative to matched WT	S211A still responds to Torin1: compatible with multisite regulation, not a failed mutant. Verify expression and exact sequence if unchanged.
S142 / combined sites	S142A changes recovery relative to WT; double mutant differs from relevant single mutants	No additional double-mutant effect: check dynamic-range ceilings and starting N/C; do not infer redundancy from nonsignificance.
mTOR-complex involvement	Independent RPTOR siRNAs reproduce the response with target/pathway engagement; competent RICTOR perturbation differs	Both perturbations act, or RICTOR depletion fails: no simple mTORC1-specific attribution. Resolve cross-talk, toxicity and RNAi specificity.
TFEB-dependent output	Both TFEB siRNAs reduce target-gene responses; WT restores them at controlled expression	Failure of rescue: verify transgene abundance/resistance; residual induction: assess incomplete depletion and TFE3/parallel pathways.
Turnover	Stimulus increases the paired inhibitor-dependent LC3-II increment; loss/rescue alters the TFEB-linked component	High LC3-II without additional inhibitor accumulation may reflect blocked clearance. Check blockade efficacy, viability and an orthogonal assay.

Claim localization only when reproducible independent-day effects agree with protein and imaging quality controls. Claim a TFEB contribution to target-gene RNA responses only after the loss-and-rescue comparison; mature RNA alone does not measure transcription rate. Claim increased operational autophagic turnover only with the paired-inhibitor result. None of these findings alone establishes direct kinase–substrate chemistry, exclusive TFEB mediation, or a universal response outside the tested HeLa conditions.

5 Discussion and Limitations

The expanded design links construct identity, nutrient-sensitive distribution, selected regulatory sites and downstream function. Its strength comes from comparisons that separate competing explanations: matched nutrient/solvent controls, endogenous confirmation, expression titration, individual and combined site substitutions, genetic perturbation, and loss-and-rescue. A planned sequence of execution is construct acceptance, cell-assay pilot, localization/recovery confirmation, then mechanism and functional modules. Failed early quality checks are resolved before downstream interpretation.

Each control has a defined scope. EGFP-only does not prove that a TFEB fusion behaves exactly like endogenous protein. Site substitutions perturb local chemistry as well as phosphorylation potential. RPTOR/RICTOR comparisons strengthen complex attribution but require competent knockdown and may reveal indirect effects; direct phosphorylation requires a separate biochemical experiment. Recovery of N/C is net redistribution, not an independently measured export rate. TFEB knockdown and rescue can support a contribution without establishing that TFEB is the only MiT/TFE-family mediator.

Nuclear accumulation, transcription and autophagic turnover are distinct endpoints. A stress response can move TFEB into nuclei without improving degradation. Drug-induced autophagy can also involve TFEB-independent pathways. Consequently, the proposed functional conclusions require their own genetic and inhibitor controls. If those assays disagree, the conclusion is narrowed to the supported level rather than describing every endpoint as "activation."

The C-terminal extension adds a question beyond reproducing redistribution. Its factorial comparison estimates genotype–fusion-architecture–stimulus effects; removing both linker and EGFP cannot assign an effect to EGFP alone. Neither alanine nor threonine substitution isolates phosphorylation from structural consequences. The primary RNA endpoint is a total effect in a calibrated expression system, while nuclear abundance and localization are possible mediators. Post-treatment adjustment cannot by itself identify intrinsic transactivation. A qualified nascent-RNA assay, TFE3 controls and comparable nuclear exposure are needed for stronger interpretation. Phosphosite and ChIP measurements have separate specificity/recovery requirements. The primary interaction may be zero despite shared effects under both drugs, so component contrasts remain essential. A null result with a wide interval is inconclusive, and this design does not establish equivalence or unprecedented novelty.

The interpretation is restricted to the stated TFEB isoform, expression range, HeLa background and drug regimens. Replication of a key contrast in an independently sourced second human cell system, with its own expression and pathway qualification, would be required before claiming generality. The revised control inventory and conditional follow-ups increase the workload beyond an 18-condition screen; the planned calendar must yield to assay qualification and the fixed precision-based sample size.

This remains an experimental proposal: no wet-lab clones, images, blots, qPCR measurements or flux data have been acquired. The new sequence files establish intended mutant identities, not their cellular activity. Supplier-specific reagents, siRNA targets, qPCR oligos, antibody performance and final doses/sample size require documented pilot validation. These implementation items and actual author contributions must be resolved before execution or course submission. The design can address major foreseeable objections; it cannot guarantee that every alternative mechanism has been excluded.

Author Contributions

The agreed division of responsibility assigns molecular cloning and construct design (part B) to Siyuan Jiang, and the biological rationale and cell-assay design (part A) to Ziyan Song. This is a planned

division, not a record of completed individual contributions. With Siyuan Jiang's authorization, AI-assisted integration also expanded part A to include explicit control, mechanism, rescue and functional-assay designs; this does not establish either author's completed personal contribution. Each author must confirm their actual contributions and review the integrated manuscript before submission. No wet-laboratory work is reported.

AI Use Declaration

AI assistance (OpenAI Codex) was used for literature checking, sequence-design code, local computational checks, drafting and integration, figure preparation, and LaTeX compilation and layout review. The circular maps were rendered with SnapGene Viewer from sequence-verified display copies; their English size-unit labels and panel layout were typeset from native SVG exports. The NPG/nrc palette follows the open-source ggsci palette inspired by Nature Reviews Cancer. The cloning and evidence-flow illustrations were generated with the built-in ImageGen tool, corrected after inspection, traced into vector paths with VTracer, and combined with editable labels and arrows. Their visual approach was informed by the Bioicons and D2 GitHub projects; no third-party icon was copied. Aligned fusion cassettes, sampling, the factorial matrix and predicted band positions were drawn by code from declared design choices and accepted sequence/fragment tables. Earlier draft illustrations remain archived. Computational checks and conceptual artwork are distinct from experimental observations. Source images, prompts, calculation inputs and reproducible scripts accompany the design. The authors remain responsible for verifying the scientific claims, practical feasibility, final parameters and contribution statement before submission.

References

[1] Stephen A. Bustin et al. "The MIQE guidelines: minimum information for publication of quantitative real-time PCR experiments". In: *Clinical Chemistry* 55.4 (2009), pp. 611–622. DOI: 10.1373/clinchem.2008.112797.

[2] Silvia Vega-Rubin-de Celis et al. "Multistep regulation of TFEB by MTORC1". In: *Autophagy* 13.3 (2017), pp. 464–472. DOI: 10.1080/15548627.2016.1271514.

[3] Clontech Laboratories. *pEGFP-N1 Vector Information, PT3027-5 / PR93634*. Manufacturer document dated 19 March 1999, accessed through a third-party mirror. 1999. URL: https://engibody.com/UPLOADEDFILES/VIAFCKEDITOR/files/AT1670-pEGFP-N1-Vector-Information.pdf (visited on 10/23/2026).

[4] Eurofins Genomics. *Eurofins sample submission guide for Sanger*. URL: https://eurofinsgenomics.com/en/products/dna-sequencing/sample-submission-guide-for-sanger/ (visited on 10/23/2026).

[5] Eurofins Genomics. *Eurofins Sanger FAQ quality scores*. URL: https://eurofinsgenomics.eu/en/eurofins-genomics/product-faqs/sanger-sequencing/ (visited on 10/23/2026).

[6] Eurofins Genomics. *Eurofins sequencing FAQ*. URL: https://eurofinsgenomics.com/en/resources/other-resources/faqs.aspx (visited on 10/23/2026).

[7] Ferguson laboratory. *pEGFP-N1-TFEB: Addgene plasmid 38119*. RRID:Addgene_38119; reference reagent record, not a claim of acquisition. Addgene. URL: https://www.addgene.org/38119/ (visited on 10/23/2026).

[8] GenBank. *Cloning vector pEGFP-N1, complete sequence, U55762.1*. URL: https://www.ncbi.nlm.nih.gov/sviewer/viewer.fcgi?id=U55762.1&db=nuccore&report=genbank&retmode=text (visited on 10/23/2026).

[9] Integrated DNA Technologies. *gBlocks and gBlocks HiFi Gene Fragments*. URL: https://www.idtdna.com/pages/products/genes-and-gene-fragments/double-stranded-dna-fragments/gblocks-gene-fragments (visited on 10/23/2026).

[10] Jose A. Martina et al. “MTORC1 functions as a transcriptional regulator of autophagy by preventing nuclear transport of TFEB”. In: *Autophagy* 8.6 (2012), pp. 903–914. DOI: 10.4161/auto.19653.

[11] Jose A. Martina et al. “The nutrient-responsive transcription factor TFE3 promotes autophagy, lysosomal biogenesis, and clearance of cellular debris”. In: *Science Signaling* 7.309 (2014), ra9. DOI: 10.1126/scisignal.2004754.

[12] Gennaro Napolitano et al. “mTOR-dependent phosphorylation controls TFEB nuclear export”. In: *Nature Communications* 9, 3312 (2018). DOI: 10.1038/s41467-018-05862-6.

[13] NCBI RefSeq. *Homo sapiens TFEB transcript variant 1, NM_007162.3*. URL: https://www.ncbi.nlm.nih.gov/sviewer/viewer.fcgi?id=NM_007162.3&db=nuccore&report=genbank&retmode=text (visited on 10/23/2026).

[14] Florentina Negoita et al. “AMPK promotes TFEB transcriptional activity through dephosphorylation at both MTORC1-dependent and -independent sites”. In: *Autophagy* 22.6 (2026), pp. 1386–1400. DOI: 10.1080/15548627.2026.2629720.

[15] New England Biolabs. *BamHI-HF R3136*. URL: https://www.neb.com/en-us/products/r3136-bamhi-hf (visited on 10/23/2026).

[16] New England Biolabs. *Cleavage close to the end of DNA fragments*. URL: https://www.neb.com/en/tools-and-resources/usage-guidelines/cleavage-close-to-the-end-of-dna-fragments (visited on 10/23/2026).

[17] New England Biolabs. *DNA ligation with T4 DNA Ligase M0202*. URL: https://www.neb.com/en/protocols/0001/01/01/dna-ligation-with-t4-dna-ligase-m0202?pdf=true (visited on 10/23/2026).

[18] New England Biolabs. *Double digest protocol*. URL: https://www.neb.com/en-gb/protocols/double-digest-protocol-with-standard-restriction-enzymes?pdf=true (visited on 10/23/2026).

[19] New England Biolabs. *Gel Loading Dye Purple (6X) B7024*. URL: https://www.neb.com/en-us/products/b7024-gel-loading-dye-purple-6x (visited on 10/23/2026).

[20] New England Biolabs. *HindIII-HF R3104*. URL: https://www.neb.com/en/products/r3104-hindiii-hf (visited on 10/23/2026).

[21] New England Biolabs. *Monarch Spin DNA Gel Extraction T1120 protocol*. URL: https://www.neb.com/en/protocols/standard-gel-extraction-protocol-using-the-monarch-spin-dna-gel-extraction-kit-and-centrifugation?pdf=true (visited on 10/23/2026).

[22] New England Biolabs. *Monarch Spin PCR and DNA Cleanup T1130 protocol*. URL: https://www.neb.com/protocols/standard-cleanup-protocol-using-the-monarch-spin-pcr-and-dna-cleanup-kit-and-centrifugation?pdf=true (visited on 10/23/2026).

[23] New England Biolabs. *Monarch Spin Plasmid Miniprep T1110*. URL: https://www.neb.com/en/protocols/plasmid-miniprep-protocol-using-centrifugation-t1110?pdf=true (visited on 10/23/2026).

[24] New England Biolabs. *Monarch T1110 buffer preparation*. URL: https://www.neb.com/en/protocols/buffer-preparation-guidance-neb-t1110 (visited on 10/23/2026).

[25] New England Biolabs. *NEB 1 kb DNA Ladder N3232*. URL: https://www.neb.com/en/products/n3232-1-kb-dna-ladder (visited on 10/23/2026).

[26] New England Biolabs. *NEB 5-alpha Competent E. coli C2987*. URL: https://www.neb.com/en-us/products/c2987-neb-5-alpha-competent-e-coli-high-efficiency (visited on 10/23/2026).

[27] New England Biolabs. *NEB 5-alpha high efficiency transformation C2987*. URL: https://www.neb.com/en-us/protocols/high-efficiency-transformation-protocol-c2987?pdf=true (visited on 10/23/2026).

[28] New England Biolabs. *OneTaq 2X Master Mix with Standard Buffer M0482 protocol*. URL: https://www.neb.com/protocols/protocol-for-onetaq-2x-master-mix-with-standard-buffer-m0482?country=en&device=pdf (visited on 10/23/2026).

[29] New England Biolabs. *Q5 High-Fidelity 2X Master Mix M0492 protocol*. URL: https://www.neb.com/en/protocols/protocol-for-q5-high-fidelity-2x-master-mix-m0492?pdf=true (visited on 10/23/2026).

[30] Mathieu Paquette et al. “AMPK-dependent phosphorylation is required for transcriptional activation of TFEB and TFE3”. In: *Autophagy* 17.12 (2021), pp. 3957–3975. DOI: 10.1080/15548627.2021.1898748.

[31] Primer3-py developers. *Primer3-py 2.4.0 thermodynamic bindings*. URL: https://libnano.github.io/primer3-py/api/bindings.html (visited on 10/23/2026).

[32] Michal Rabani et al. "Metabolic labeling of RNA uncovers principles of RNA production and degradation dynamics in mammalian cells". In: *Nature Biotechnology* 29.5 (2011), pp. 436–442. DOI: 10.1038/nbt.1861.

[33] Agnes Roczniak-Ferguson et al. "The transcription factor TFEB links mTORC1 signaling to transcriptional control of lysosome homeostasis". In: *Science Signaling* 5.228 (2012), ra42. DOI: 10.1126/scisignal.2002790.

[34] Carmine Settembre et al. "A lysosome-to-nucleus signalling mechanism senses and regulates the lysosome via mTOR and TFEB". In: *The EMBO Journal* 31.5 (2012), pp. 1095–1108. DOI: 10.1038/emboj.2012.32.

[35] Carmine Settembre et al. "TFEB links autophagy to lysosomal biogenesis". In: *Science* 332.6036 (2011), pp. 1429–1433. DOI: 10.1126/science.1204592.